

## Unit 8

### Partial Differential Equation of the Second Order

A partial differential equation in which partial differential coefficients of  $z$  occur with respect to  $x$  and  $y$  only present in the equation or  $z$  in one is called they linear PDE. The general form of linear PDE is

$$[A_0 \frac{\partial^n z}{\partial x^n} + A_1 \frac{\partial^n z}{\partial x^{n-1} \partial y} + A_2 \frac{\partial^n z}{\partial x^{n-2} \partial y^2} + \dots + A_n \frac{\partial^n z}{\partial y^n} + [B_0 \frac{\partial^{n-1} z}{\partial x^{n-1}} + B_1 \frac{\partial^{n-1} z}{\partial x^{n-2} \partial y} + \dots + B_{n-1} \frac{\partial^{n-1} z}{\partial y^{n-1}}] + \dots + [K_0 \frac{\partial z}{\partial x} + K_1 \frac{\partial z}{\partial y}] + Lz = f(x, y) \quad \dots \quad (1)$$

where  $A_0, A_1, \dots, A_n; B_0, B_1, \dots, B_n; \dots; K_0, K_1, L$  are the coefficients (constants or variables).

The operations  $\frac{\partial}{\partial x}, \frac{\partial}{\partial y}$  are denoted by  $D$  and  $D'$  respectively i.e.  $\frac{\partial}{\partial x} = D$  and  $\frac{\partial}{\partial y} = D'$ . Then, the equation (1) can be written as

$$[A_0 D^n + A_1 D^{n-1} D' + A_2 D^{n-2} D'^2 + \dots + A_n D'^n]z + [B_0 D^{n-1} + B_1 D^{n-2} D' + \dots + B_n D'^{n-1}]z + \dots + [K_0 D + K_1 D']z + Lz = f(x, y) \quad \dots \quad (2)$$

More formally, it is written as

$$F(D, D')z = f(x, y) \quad \dots \quad (3)$$

Where

$$F(D, D') = [A_0 D^n + A_1 D^{n-1} D' + A_2 D^{n-2} D'^2 + \dots + A_n D'^n] + [B_0 D^{n-1} + B_1 D^{n-2} D' + \dots + B_n D'^{n-1}] + \dots + [K_0 D + K_1 D'] + L.$$

A linear PDE of second order is of the form

$$A_0 \frac{\partial^2 z}{\partial x^2} + A_1 \frac{\partial^2 z}{\partial x \partial y} + A_2 \frac{\partial^2 z}{\partial y^2} + A_3 \frac{\partial z}{\partial x} + A_4 \frac{\partial z}{\partial y} + Lz = f(x, y)$$

If the coefficient  $A_0, A_1, A_2, A_3, A_4, L$  are all constants, then this equation is called the linear PDE of order two with constant coefficients.

The most general form of the second order PDE is

$$f(x, y, z, p, q, r, s, t) = 0$$

where  $p, q, r, s, t$  have their users meaning.

Examples

$$(i) r + 2s + t = 0$$

$$(ii) x^2 r - 2xz + t + q = 0$$

Note

- i. If at least one of the coefficients  $A_0, A_1, \dots, A_n; B_0, B_1, \dots, B_n; \dots; K_0, K_1, L$  is not constant then the equation is called linear PDE with variable coefficients.
- ii. If  $f(x, y) = 0$  then, the equation is called the homogeneous PDE.
- iii. Linear PDE of second order

$$A_0 \frac{\partial^2 z}{\partial x^2} + A_1 \frac{\partial^2 z}{\partial x \partial y} + A_2 \frac{\partial^2 z}{\partial y^2} + A_3 \frac{\partial z}{\partial x} + A_4 \frac{\partial z}{\partial y} + Lz = f(x, y)$$

$$(A_0 D^2 + A_1 D D' + A_2 D'^2 + A_3 D + A_4 D' + L)z = f(x, y).$$

#### Origin of Second Order PDE

Let  $z$  be a function of the type

$$z = f(u) + g(v) + w \quad \dots \quad (1)$$

where,  $f$  and  $g$  and arbitrary functions of  $u$  and  $v$  respectively.

Also,  $u, v, w$  are given function of  $x$  and  $y$ . Then,

$$p = \frac{\partial z}{\partial x}, q = \frac{\partial z}{\partial y}, r = \frac{\partial p}{\partial x} = \frac{\partial^2 z}{\partial x^2}, s = \frac{\partial p}{\partial y} = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial q}{\partial x}, t = \frac{\partial q}{\partial y} = \frac{\partial^2 z}{\partial y^2} \quad \dots \quad (2)$$

Differentiating (1) w.r.t  $x$  and  $y$ , we get

$$p = f'(u)u_x + g'(v)v_x + w_x \quad \dots \quad (3)$$

$$q = f'(u)u_y + g'(v)v_y + w_y \quad \dots \quad (4)$$

and

$$r = f(u)u_x^2 + g(v)v_x^2 + f'(u)u_{xx} + g'(v)v_{xx} + w_{xx} \quad \dots \quad (5)$$

$$s = f(u)u_x u_y + g(v)v_x v_y + f'(u)u_{xy} + g'(v)v_{xy} + w_{xy} \quad \dots \quad (6)$$

$$t = f(u)u_y^2 + g(v)v_y^2 + f'(u)u_{yy} + g'(v)v_{yy} + w_{yy} \quad \dots \quad (7)$$

Now, eliminating  $f', f'', g', g''$  from five equations (3) - (7), we get

$$\begin{vmatrix} p - w_x & u_x & v_x & 0 & 0 \\ q - w_y & u_y & v_y & 0 & 0 \\ r - w_{xx} & u_{xx} & v_{xx} & u_x^2 & v_x^2 \\ s - w_{xy} & u_{xy} & v_{xy} & u_x u_y & v_x v_y \\ t - w_{yy} & u_{yy} & v_{yy} & u_y^2 & v_y^2 \end{vmatrix} \quad \dots \quad (8)$$

Which involves only partial derivatives  $p, q, r, s, t$ , and the known functions of  $x$  and  $y$ . Therefore, it is a PDE of second order. If this determinant is expanded, an equation of the form

$$Rr + Ss + Tt + Pp + Qq = W \quad \dots \quad (9)$$

is obtained, where  $R, S, T, P, Q, W$  are known functions of  $x$  and  $y$ .

Hence, the relation (1) is a solution of the second order linear PDE (9). We also note that equation (9) does not involve the dependent variable  $z$ .

### Example 1

Find a differential equation from  $z = f(x + ay) + g(x - ay)$  by eliminating the arbitrary functions  $f$  and  $g$ , where  $a$  is some constant.

### Solution

The given equation is

$$z = f(x + ay) + g(x - ay) \quad \dots \quad (1)$$

Differentiating (1) partially w.r.t  $x$  and  $y$ , we get

$$p = \frac{\partial z}{\partial x} = f'(x + ay) + g'(x - ay) \quad \dots \quad (2)$$

$$q = \frac{\partial z}{\partial y} = af'(x + ay) - ag'(x - ay) \quad \dots \quad (3)$$

Again, differentiating (2) partially w.r.t.  $x$ , we get

$$r = \frac{\partial^2 z}{\partial x^2} = f''(x + ay) + g''(x - ay) \quad \dots \quad (4)$$

And differentiating (3) partially w.r.t.  $y$  we get

$$t = \frac{\partial^2 z}{\partial y^2} = a^2 f''(x + ay) + a^2 g''(x - ay) \quad \dots \quad (5)$$

$$\Rightarrow t = a^2 [f''(x + ay) + g''(x - ay)]$$

$$t = a^2 r, \text{ using (4)}$$

The required differential equation is

$$t = a^2 r.$$

### Homogeneous Linear PDEs With Constant Coefficients

A PDE in which the dependent variable and its derivatives, occur only in the first degree and are not multiplied together, their coefficients being constants or functions of  $x$  and  $y$  is known as a linear PDE. The general form of such an equation is:

$$[A_0 \frac{\partial^n z}{\partial x^n} + A_1 \frac{\partial^n z}{\partial x^{n-1} \partial y} + A_2 \frac{\partial^n z}{\partial x^{n-2} \partial y^2} + \dots + A_n \frac{\partial^n z}{\partial y^n} + [B_0 \frac{\partial^{n-1} z}{\partial x^{n-1}} + B_1 \frac{\partial^{n-1} z}{\partial x^{n-2} \partial y} + \dots + B_{n-1} \frac{\partial^{n-1} z}{\partial y^{n-1}}] + \dots + [M_0 \frac{\partial z}{\partial x} + B_1 \frac{\partial z}{\partial y}] + N_0 z = f(x, y) \quad \dots \quad (1)$$

where the coefficients  $A_0, A_1, \dots, A_n; B_0, B_1, \dots, B_{n-1}; \dots; M_0, M_1, N_0$  are constants or functions of  $x$  and  $y$ . If all these coefficients are constants, then (1) is called a linear PDE with constant and coefficients.

For convenience  $\frac{\partial}{\partial x}$  and  $\frac{\partial}{\partial y}$  are denoted by  $D$  and  $D'$  respectively then (1) reduces to

$$[A_0 D^n + A_1 D^{n-1} D' + A_2 D^{n-2} D'^2 + \dots + A_n D'^n]z + [B_0 D^{n-1} + B_1 D^{n-2} D' + \dots + B_{n-1} D'^{n-1}]z + \dots + [M_0 D + M_1 D']z + N_0 z = f(x, y) \quad \dots \quad (2)$$

Or

$$F(D, D')z = f(x, y) \quad \dots \quad (3)$$

If all the derivatives present in the equation (1) are of the same order, then it is called a linear homogeneous PDE with constant coefficients and is of the form

$$[A_0 D^n + A_1 D^{n-1} D' + A_2 D^{n-2} D'^2 + \dots + A_n D'^n]z = f(x, y) \quad \dots \quad (4)$$

On the other hand, it is called non-homogeneous if the orders of all the derivatives are not equal.

### **Solution of Homogeneous Linear PDE With Constant Coefficients**

The homogeneous linear PDE with constant coefficients is

$$[A_0 D^n + A_1 D^{n-1} D' + A_2 D^{n-2} D'^2 + \dots + A_n D'^n]z = f(x, y) \quad \dots \quad (1)$$

where the coefficient  $A_0, A_1, A_2, \dots, A_n$  are constants and can be written as

$$F(D, D')z = f(x, y) \quad \dots \quad (2)$$

#### **Theorem 1**

If  $u$  is the complementary function and  $z'$  is a particular integral of a linear PDE

$$F(D, D')z = f(x, y),$$

then  $u + z'$  is a general solution of the equation.

#### **Proof**

The given equation is

$$F(D, D')z = f(x, y) \quad \dots \quad (1)$$

The complementary function  $u$  of (1) is the general solution of

$$F(D, D')z = 0 \quad \dots \quad (2)$$

$$\therefore F(D, D')u = 0 \quad \dots \quad (3)$$

We know that the complementary function must contain as many arbitrary constants as the order of the equation (2).

Any solution  $z'$  of (1) is called a particular integral of (1). We know that particular solution does not contain any arbitrary constants. Thus, by definition, we have

$$F(D, D')z' = f(x, y) \quad \dots \quad (4)$$

Adding (3) and (4), we get

$$\begin{aligned} F(D, D')u + F(D, D')z' &= 0 + f(x, y) \\ \Rightarrow F(D, D')(u + z') &= f(x, y) \end{aligned}$$

which shows that  $u + z$  is a solution of

$$F(D, D')z = f(x, y).$$

Since (1) and (2) are of the same order, the general solution  $u + z'$  will contain as many arbitrary constants as the general solution of (1) requires.

This completes the proof of the theorem.

#### **Theorem 2**

If  $u_1, u_2, u_3, \dots, u_n$  are solutions of homogeneous linear PDE  $F(D, D')z = 0$ , then  $\sum_{r=1}^n c_r u_r$  is also a solution, where  $c_1, c_2, \dots, c_n$  are arbitrary constants.

**Proof**

The given equation is

$$F(D, D')z = 0 \quad \dots \quad (1)$$

We know that

$$F(D, D')(c_r u_r) = c_r F(D, D')u_r \quad \dots \quad (2)$$

and also,

$$F(D, D') \sum_{r=1}^n u_r = \sum_{r=1}^n F(D, D')u_r \quad \dots \quad (3)$$

for any functions  $u_r$ .

Then, from (2) and (3) we have

$$F(D, D') \sum_{r=1}^n (c_r u_r) = \sum_{r=1}^n F(D, D')(c_r u_r) = c_r \sum_{r=1}^n F(D, D')u_r \quad \dots \quad (4)$$

since  $u_r, r = 1, 2, \dots, n$  is a solution of  $F(D, D')z = 0$ , we have,

$$F(D, D')u_r = 0 \text{ for all } r = 1, 2, \dots, n$$

(4) implies

$$F(D, D') \sum_{r=1}^n (c_r u_r) = 0.$$

This proves the theorem.

**Method of Finding Complementary Function (C.F.) of Homogeneous Linear PDE with Constant Coefficients.**

The homogeneous linear PDE with constant coefficient is

$$F(D, D')z = f(x, y)$$

that is,

$$[A_0 D^n + A_1 D^{n-1} D' + A_2 D^{n-2} D'^2 + \dots + A_n D'^n]z = f(x, y) \quad \dots \quad (1)$$

where,  $A_0, A_1, A_2, \dots, A_n$  are constants.

The complementary function (C.F.) of (1) is the general solution of

$$[A_0 D^n + A_1 D^{n-1} D' + A_2 D^{n-2} D'^2 + \dots + A_n D'^n]z = 0 \quad \dots \quad (2)$$

$$\text{or, } [(D - m_1 D')(D - m_2 D') \dots (D - m_n D')]z = 0 \quad \dots \quad (3)$$

where,  $m_1, m_2, \dots, m_n$  are some constants.

Clearly, the solution of any one of the equations

$$(D - m_1 D')z = 0, (D - m_2 D')z = 0, \dots (D - m_n D')z = 0 \quad \dots \quad (4)$$

is also a solution of (3).

We now show that the general solution of  $(D - mD')z = 0$  is

$z = \phi(y + mx)$  where  $\phi$  is an arbitrary function.

Now,

$$\begin{aligned} (D - mD')z &= 0 \\ \Rightarrow \frac{\partial z}{\partial x} - m \frac{\partial z}{\partial y} &= 0 \\ \Rightarrow p - mq &= 0 \quad \dots \quad (5) \end{aligned}$$

Which is of the form

$$Pp + Qq = R$$

So, Lagrange's A.E. of (5) are

$$\frac{dx}{1} = \frac{dy}{-m} = \frac{dz}{0} \quad \dots \quad (6)$$

From first two ratios

$$\frac{dx}{1} = \frac{dy}{-m} \Rightarrow dy + m dx = 0.$$

Integration implies

$$y + mx + c_1 \quad \dots \quad (7)$$

From third ratio, we have

$$dz = 0 \Rightarrow z = c_2 \quad \dots \quad (8)$$

Therefore, from (7) and (8), the general solution of (5) is

$$z = \phi(y + mx)$$

where  $\phi$  is an arbitrary function.

So, we assume that a solution of (2) is of the form

$$z = \phi(y + mx) \quad \dots \quad (9)$$

From (9)

$$\begin{aligned} Dz &= \frac{\partial z}{\partial x} = m\phi'(y + mx) \\ D^2z &= \frac{\partial^2 z}{\partial x^2} = m^2\phi''(y + mx) \\ &\dots \dots \dots \dots \\ &\dots \dots \dots \dots \\ D^n z &= \frac{\partial^n z}{\partial x^n} = m^n \phi^n(y + mx) \end{aligned}$$

Again,

$$\begin{aligned} D'z &= \frac{\partial z}{\partial y} = \phi'(y + mx) \\ D'^2 z &= \frac{\partial^2 z}{\partial y^2} = \phi''(y + mx) \\ &\dots \dots \dots \dots \\ &\dots \dots \dots \dots \\ D'^n z &= \frac{\partial^n z}{\partial y^n} = \phi^n(y + mx). \end{aligned}$$

Also,

$$D^r D'^s z = \frac{\partial^{r+s} z}{\partial x^r \partial y^s} = m^r \phi^{r+s}(y + mx).$$

Substituting these values in (2) and simplifying we get

$$[A_0 m^n + A_1 m^{n-1} + A_2 m^{n-2} + \dots + A_n] \phi^n(y + mx) = 0$$

which is true when  $m$  is a root of the equation

$$A_0 m^n + A_1 m^{n-1} + A_2 m^{n-2} + \dots + A_n = 0 \quad \dots \quad (10)$$

The equation (10) is called the auxiliary equation and is obtained by putting  $D = m$  and  $D' = 1$  in  $F(D, D') = 0$ .

Let  $m_1, m_2, \dots, m_n$  be  $n$  roots of (10). Then, two cases arise.

**Case I [When  $m_1, m_2, \dots, m_n$  are all Distinct]**

Then the part of C.F. corresponding to  $m = m_r$  is  $z = \phi_r(y + m_r x)$  for  $r = 1, 2, \dots, n$ . Since (2) is linear, the sum of the solutions is also a solution. Therefore, the C.F. of (1) is

$$z = \phi_1(y + m_1 x) + \phi_2(y + m_2 x) + \dots + \phi_n(y + m_n x) \quad \dots \quad (11)$$

where  $\phi_1, \phi_2, \phi_3 \dots, \phi_n$  are arbitrary functions.

### Case II [Repeated Roots]

If a root  $m$  is repeated for  $r$  times then, the C.F. is given by

$$z = \phi_1(y + mx) + x\phi_2(y + mx) + x^2\phi_3(y + mx) + \dots + x^{r-1}\phi_r(y + mx) + \phi_{r+1}(y + m_{r+1}x) \dots + \phi_n(y + m_nx) \dots \quad (11)$$

where  $\phi_1, \phi_2, \phi_3 \dots, \phi_n$  are arbitrary functions.

### Working Rules of Finding C.F.

Let the given PDE be  $F(D, D')z = f(x, y)$ . Factorize  $F(D, D')$  into linear factors of the form  $(bD - aD')$ . Then, use the following results.

- Corresponding to each non-repeated factor  $(bD - aD')$ , the part of C.F. is taken as  $\phi(by + ax)$ .
- Corresponding to a repeated factor  $(bD - aD')^r$ , the part of C.F. is taken as  $\phi_1(y + mx) + x\phi_2(y + mx) + \dots + x^{r-1}\phi_r(y + mx)$ .
- Corresponding to a non-repeated factor  $D$ , part of C.F. is taken as  $\phi(y)$ .
- Corresponding to a repeated factor  $D^r$ , the part of C.F. is taken as  $\phi_1(y) + x\phi_2(y) + \dots + x^{r-1}\phi_r(y)$ .
- Corresponding to a non-repeated factor  $D'$ , the part of C.F. is taken as  $\phi(x)$ .
- Corresponding to a repeated factor  $(D')^r$ , the part of C.F. is taken as  $\phi_1(x) + y\phi_2(x) + \dots + y^{r-1}\phi_r(x)$ .

### Examples

1. Solve  $r = a^2t$

#### Solution

The given PDE is

$$\begin{aligned} r &= a^2t \\ \frac{\partial^2 z}{\partial x^2} &= a^2 \frac{\partial^2 z}{\partial y^2} \\ (D^2 - a^2D'^2)z &= 0 \quad \dots \quad (1) \end{aligned}$$

The Auxiliary Equation of (1) is

$$\begin{aligned} m^2 - a^2 &= 0 \\ m &= a, -a \end{aligned}$$

Therefore, the C.F. of (1) is

$$z = \phi_1(y + ax) + \phi_2(y - ax).$$

Hence, the general solution is

$$z = \phi_1(y + ax) + \phi_2(y - ax)$$

where  $\phi_1, \phi_2$  are arbitrary functions.

2. Solve  $2r + 5s + 2t = 0$

#### Solution

The given PDE is

$$\begin{aligned} 2r + 5s + 2t &= 0 \\ \Rightarrow 2 \frac{\partial^2 z}{\partial x^2} + 5 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} &= 0 \quad \dots \quad (1) \end{aligned}$$

It's Auxiliary equation is

$$\begin{aligned} 2m^2 + 5m + 2 &= 0 \\ \Rightarrow (2m + 1)(m + 2) &= 0 \\ \therefore m &= -\frac{1}{2}, -2 \end{aligned}$$

Therefore, the C.F. of (1) is

$$z = \phi_1(2y - x) + \phi_2(y - 2x)$$

Hence, the general solution of (1) is

$$z = \phi_1(2y - x) + \phi_2(y - 2x)$$

where  $\phi_1$  and  $\phi_2$  are arbitrary functions.

3. Solve  $(D^3 - 4D^2D' + 4DD'^2)z = 0$

**Solution**

The given PDE is

$$\begin{aligned}(D^3 - 4D^2D' + 4DD'^2)z &= 0 \\ \Rightarrow D(D^2 - 4DD' + 4D'^2)z &= 0 \\ \Rightarrow D(D - 2D')^2z &= 0\end{aligned}$$

The C.F. of (1) is

$$z = \phi_1(y) + \phi_2(y + 2x) + x\phi_3(y + 2x)$$

where  $\phi_1, \phi_2, \phi_3$  are arbitrary functions.

4. Solve  $(D^3D'^2 + D^2D'^3)z = 0$

**Solution**

The given PDE is

$$\begin{aligned}(D^3D'^2 + D^2D'^3)z &= 0 \\ \Rightarrow D^2D'^2(D + D')z &= 0 \quad \dots \quad (1)\end{aligned}$$

Therefore, the C.F. of (1) is

$$z = \phi_1(y) + x\phi_2(y) + \phi_3(x) + y\phi_4(x) + \phi_5(y - x)$$

where,  $\phi_1, \phi_2, \phi_3, \phi_4, \phi_5$  are arbitrary functions.

**Particular Integral (P.I.) of Homogeneous Linear PDE  $F(D, D')z = f(x, y)$**

The inverse operator  $\frac{1}{F(D, D')}$  of the operator  $F(D, D')$  is defined by the following identity.

$$F(D, D')\left[\frac{1}{F(D, D')}f(x, y)\right] = f(x, y).$$

Therefore, particular integral P.I. of  $F(D, D')z = f(x, y)$  is

$$\frac{1}{F(D, D')}f(x, y).$$

**Note**

i.  $D, D^2, \dots$  will stand for differentiating partially w.r.t.  $x$  once, twice and so on.

For example

$$\begin{aligned}Dx^4y^5 &= \frac{\partial}{\partial x}(x^4y^5) = 4x^3y^5 \\ D^2x^4y^5 &= \frac{\partial^2}{\partial x^2}(x^4y^5) = 12x^2y^5\end{aligned}$$

ii.  $D', D'^2, \dots$  stand for differentiating partially w.r.t.  $y$  once twice and so on.

$$\begin{aligned}D'x^4y^5 &= \frac{\partial}{\partial y}(x^4y^5) = 5x^4y^4 \\ D'^2x^4y^5 &= \frac{\partial^2}{\partial y^2}(x^4y^5) = 20x^4y^3\end{aligned}$$

iii.  $\frac{1}{D}, \frac{1}{D^2}, \dots$  stand for integrating partially w.r.t.  $x$  once, twice and so on.

$$\frac{1}{D}x^4y^5 = \int x^4y^5 dx = \frac{x^5y^5}{5}$$

$$\frac{1}{D^2} x^4 y^5 = \int \int x^4 y^5 dx dx = \frac{x^6 y^5}{30}$$

iv.  $\frac{1}{D'}, \frac{1}{D'^2}$  ... stand for integrating partially w.r.t.  $y$  once, twice and so on.

$$\frac{1}{D'} x^4 y^5 = \int x^4 y^5 dy = \frac{x^4 y^6}{6}$$

$$\frac{1}{D'^2} x^4 y^5 = \int \int x^4 y^5 dy dy = \frac{x^4 y^7}{42}$$

### Short Methods of Finding P.I.

**Method I : When  $f(x, y)$  is of the form  $(ax + by)$**

#### Theorem

If  $f(D, D')$  is a homogeneous function of  $D$  and  $D'$  of degree  $n$ , then

$$\begin{aligned} P.I. &= \frac{1}{F(D, D')} \phi^n(ax + by) \\ &= \frac{1}{F(D, D')} \phi(ax + by) \end{aligned}$$

provided  $F(a, b) \neq 0$ ,  $\phi^n$  being the  $n^{\text{th}}$  derivative of  $\phi$  with respect to  $ax + by$  as a whole.

#### Working Rule

If  $F(a, b) \neq 0$ , then

$$P.I. = \frac{1}{F(D, D')} \phi^n(ax + by) = \frac{1}{F(a, b)} \int \int \dots \int f(v) dv dv \dots dv$$

where  $v = ax + by$ . Note that the RHS contains a multiple integral of  $n^{\text{th}}$  order.

#### Theorem 2

$F(a, b) = 0$  if and only if  $(bD - aD')$  is a factor of  $F(D, D')$ . Then, when  $F(a, b) = 0$ ,

$$P.I. = \frac{1}{(bD - aD')^n} \phi(ax + by) = \frac{x^n}{b^n n!} \phi(ax + by).$$

#### Examples

1. Solve  $(D^2 + 3DD' + 2D'^2)z = x + y$

#### Solution

The given equation is

$$(D^2 + 3DD' + 2D'^2)z = x + y \quad \dots \quad (1)$$

The Auxiliary equation is

$$\begin{aligned} m^2 + 3m + 2 &= 0 \\ \Rightarrow (m + 2)(m + 1) &= 0 \\ \therefore m &= -1, -2 \end{aligned}$$

$$\therefore C.F. = \phi_1(y - x) + \phi_2(y - 2x),$$

where  $\phi_1$  and  $\phi_2$  are arbitrary functions.

Next,

Particular integral  $P.I. = \frac{1}{F(D, D')} (x + y)$

$$\begin{aligned} &= \frac{1}{D^2 + 3DD' + 2D'^2} (x + y) \\ &= \frac{1}{1^2 + 3.1.1 + 2.1^2} \iint v dv dv, \quad \text{where } v = x + y \\ &= \frac{1}{6} \int \frac{v^2}{2} dv \end{aligned}$$



$$= \frac{1}{6} \cdot \frac{v^3}{6} = \frac{(x+y)^3}{36}$$

$$\therefore P.I = \frac{(x+y)^3}{36}.$$

Hence, the general solution is

$$z = C.F. + P.I.$$

$$z = \phi_1(y-x) + \phi_2(y-2x) + \frac{(x+y)^3}{36}$$

2. Solve  $(D^3 - 6D^2D' + 11DD'^2 - 6D'^3)z = e^{5x+6y}$ .

**Solution**

The given PDE is

$$(D^3 - 6D^2D' + 11DD'^2 - 6D'^3)z = e^{5x+6y} \quad \dots (1)$$

It's Auxiliary equation is

$$m^3 - 6m^2 + 11m - 6 = 0$$

Solving we get  $m = 1, 2, 3$ .

$$\therefore C.F. = \phi_1(y+x) + \phi_2(y+2x) + \phi_3(y+3x)$$

where  $\phi_1, \phi_2, \phi_3$  are arbitrary functions.

**For P.I.**

$$\begin{aligned} P.I. &= \frac{1}{F(D, D')} e^{5x+6y} \\ &= \frac{1}{D^3 - 6D^2D' + 11DD'^2 - 6D'^3} e^{5x+6y} \\ &= \frac{1}{(D-D')(D-2D')(D-3D')} e^{5x+6y} \\ &= \frac{1}{(5-6)(5-2.6)(5-3.6)} \int \int \int e^v dv dv dv, \quad \text{where } v = 5x + 6y \\ &= \frac{1}{-91} \int \int e^v dv dv, \\ &= \frac{1}{-91} \int e^v dv, \\ &= \frac{1}{-91} e^v = \frac{1}{-91} e^{5x+6y}, \\ \therefore P.I. &= \frac{1}{-91} e^{5x+6y} \end{aligned}$$

Hence, the required general solution is

$$z = C.F. + P.I.$$

That is,

$$z = \phi_1(y+x) + \phi_2(y+2x) + \phi_3(y+3x) - \frac{1}{91} e^{5x+6y}$$

3. Solve  $(D^2 - 2DD' + D'^2)z = \tan(x+y)$

**Solution**

The given PDE is

$$(D^2 - 2DD' + D'^2)z = \tan(x+y) \quad \dots (1)$$

It's A.E. is

$$m^2 - 2m + 1 = 0$$

$$\Rightarrow (m - 1)^2 = 0$$

$$\therefore m = 1, 1,$$

$$\therefore C.F. = \phi_1(y + x) + x\phi_2(y + x)$$

where  $\phi_1, \phi_2$  are arbitrary functions

**To find P.I.**

$$\begin{aligned} P.I. &= \frac{1}{F(D, D')} \tan(y + x) \\ &= \frac{1}{D^2 - 2DD' + D'^2} \tan(y + x) \\ &= \frac{1}{(D - D')^2} \tan(y + x) \\ &= \frac{x^2}{1^2 \cdot 2!} \tan(y + x) \\ &= \frac{x^2}{2} \tan(y + x) \end{aligned}$$

Therefore, the required general solution is

$$z = \phi_1(y + x) + x\phi_2(y + x) + \frac{x^2}{2} \tan(y + x)$$

4. Solve  $(D^2 - 5DD' + 4D'^2)z = \sin(4x + y)$

**Solution**

The given PDE is

$$(D^2 - 5DD' + 4D'^2)z = \sin(4x + y) \quad \dots \quad (1)$$

It's A.E. is

$$m^2 - 5m + 4 = 0$$

$$\Rightarrow m^2 - 4m - m + 4 = 0$$

$$\Rightarrow (m - 4)(m - 1) = 0$$

$$\therefore m = 1, 4$$

$$\therefore C.F. = \phi_1(y + x) + \phi_2(y + 4x)$$

where  $\phi_1, \phi_2$  are arbitrary functions.

**To find P.I.**

$$\begin{aligned} P.I. &= \frac{1}{F(D, D')} \sin(4x + y) \\ &= \frac{1}{D^2 - 5DD' + 4D'^2} \sin(4x + y) \\ &= \frac{1}{(D - 4D')(D - D')} \sin(4x + y) \\ &= \frac{1}{(D - 4D')} \left[ \frac{1}{(D - D')} \sin(4x + y) \right] \\ &= \frac{1}{(D - 4D')} \left[ \frac{1}{(4 - 1)} \int \sin(4x + y) d(4x + y) \right] \\ &= \frac{1}{(D - 4D')} \left[ -\frac{1}{3} \cos(4x + y) \right] \\ &= -\frac{1}{3} \left[ \frac{1}{(D - 4D')} \cos(4x + y) \right] \end{aligned}$$

$$\begin{aligned}
&= -\frac{1}{3} \left[ \frac{x}{1.1!} \cos(4x + y) \right] \\
&= -\frac{1}{3} \left[ \frac{x}{1.1!} \cos(4x + y) \right] \\
&= -\frac{x}{3} \cos(4x + y)
\end{aligned}$$

Therefore, the required general solution is

$$z = C.F. + P.I.$$

$$z = \phi_1(y + x) + \phi_2(y + 4x) - \frac{x}{3} \cos(4x + y)$$

5. Solve  $(D^3 - 4D^2D' + 4DD'^2)z = 4\sin(2x + y)$

**Solution**

The given PDE is

$$(D^3 - 4D^2D' + 4DD'^2)z = 4\sin(2x + y) \quad \dots \quad (1)$$

It's A.E. is

$$\begin{aligned}
m^3 - 4m^2 + 4m &= 0 \\
\text{or, } m(m^2 - 4m + 4) &= 0 \\
\text{or, } m(m - 2)^2 &= 0 \\
\therefore m &= 0, 2, 2
\end{aligned}$$

$$\therefore C.F. = \phi_1(y) + \phi_2(y + 2x) + x\phi_3(y + 2x)$$

where  $\phi_1, \phi_2, \phi_3$  are arbitrary functions.

**To find P.I.**

$$\begin{aligned}
P.I. &= \frac{1}{F(D, D')} 4\sin(2x + y) \\
&= \frac{1}{D^3 - 4D^2D' + 4DD'^2} 4\sin(2x + y) \\
&= \frac{1}{D(D - 2D')^2} 4\sin(2x + y) \\
&= 4 \frac{1}{(D - 2D')^2} \left[ \frac{1}{D} \sin(2x + y) \right] \\
&= 4 \frac{1}{(D - 2D')^2} \left[ -\frac{1}{2} \cos(2x + y) \right] \\
&= -2 \left[ \frac{1}{(D - 2D')^2} \cos(2x + y) \right] \\
&= -2 \left[ \frac{x^2}{1^2 \cdot 2!} \cos(2x + y) \right] \\
&= -x^2 \cos(2x + y)
\end{aligned}$$

Therefore,

$$P.I. = -x^2 \cos(2x + y)$$

Hence, the required general solution is

$$z = C.F. + P.I.$$

That is,

$$z = \phi_1(y) + \phi_2(y + 2x) + x\phi_3(y + 2x) - x^2 \cos(2x + y).$$

**Method II:**

When  $f(x, y)$  is of the form  $x^m y^n$  or a rational integral algebraic function of  $x$  and  $y$ .

**Example**

Solve  $(D^3 - D'^3)z = x^3 y^3$

### Solution

The given PDE is

$$(D^3 - D'^3)z = x^3y^3 \quad \dots \quad (1)$$

It's A.E. is

$$m^3 - 1 = 0 \Rightarrow (m - 1)(m^2 + m + 1) = 0$$

$$\therefore m = 1, \omega, \omega^2$$

$$\therefore C.F. = \phi_1(y + x) + \phi_2(y + \omega x) + \phi_3(y + \omega^2 x)$$

where  $\phi_1, \phi_2, \phi_3$  are arbitrary functions.

### To find P.I.

$$\begin{aligned}
P.I. &= \frac{1}{F(D, D')} x^3 y^3 \\
&= \frac{1}{(D^3 - D'^3)} x^3 y^3 \\
&= \frac{1}{D^3 \left(1 - \frac{D'^3}{D^3}\right)} x^3 y^3 \\
&= \frac{1}{D^3} \left(1 - \frac{D'^3}{D^3}\right)^{-1} x^3 y^3 \\
&= \frac{1}{D^3} \left(1 + \frac{D'^3}{D^3} + \dots\right) x^3 y^3 \\
&= \frac{1}{D^3} \left(x^3 y^3 + \frac{1}{D^3} 6x^3\right) \\
&= \frac{1}{D^3} \left(x^3 y^3 + 6 \times \frac{x^6}{4.5.6}\right) \\
&= \frac{1}{D^3} \left(x^3 y^3 + \frac{x^6}{20}\right) \\
&= \left(\frac{x^6 y^3}{4.5.6} + \frac{x^9}{20.7.8.9}\right) \\
&= \frac{x^6 y^3}{120} + \frac{x^9}{10080}
\end{aligned}$$

Hence, the required general solution is

$$z = \phi_1(y + x) + \phi_2(y + \omega x) + \phi_3(y + \omega^2 x) + \frac{x^6 y^3}{120} + \frac{x^9}{10080}$$

**Example** Solve  $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = -4\pi(x^2 + y^2)$

### Solution

The given PDE is

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = -4\pi(x^2 + y^2) \quad \dots \quad (1)$$

It's A.E. is

$$m^2 + 1 = 0 \Rightarrow m = i, -i$$

$$\therefore C.F. = \phi_1(y + ix) + \phi_2(y - ix)$$

where  $\phi_1, \phi_2$  are arbitrary functions.

To find P.I.

$$\begin{aligned}
P.I. &= \frac{1}{F(D, D')} \times [-4\pi(x^2 + y^2)] \\
&= \frac{1}{D^2 + D'^2} [-4\pi(x^2 + y^2)] \\
&= -4\pi \frac{1}{D^2 + D'^2} (x^2 + y^2) \\
&= -4\pi \times \frac{1}{D^2} \frac{1}{\left(1 + \frac{D'^2}{D^2}\right)} (x^2 + y^2) \\
&= -4\pi \times \frac{1}{D^2} \left(1 + \frac{D'^2}{D^2}\right)^{-1} (x^2 + y^2) \\
&= -4\pi \times \frac{1}{D^2} \left(1 - \frac{D'^2}{D^2} + \dots\right) (x^2 + y^2) \\
&= -4\pi \times \frac{1}{D^2} \left(x^2 + y^2 - \frac{2}{D^2}\right) \\
&= -4\pi \times \frac{1}{D^2} (x^2 + y^2 - x^2) \\
&= -4\pi \times \frac{1}{D^2} (y^2) \\
&= -4\pi \times \frac{x^2 y^2}{2} \\
&= -2\pi x^2 y^2
\end{aligned}$$

Hence, the required general solution is

$$z = \phi_1(y + ix) + \phi_2(y - ix) - 2\pi x^2 y^2$$

### Example

Solve  $(D^2 - 2DD' + D'^2)z = e^{x+2y} + x^3$ .

### Nonhomogeneous Linear PDE With Constant Coefficients

#### Definition

A linear PDE with constant coefficients is known as non-homogeneous linear PDE with constant coefficients if the orders of all the partial derivatives involved in the equation are not equal. For example

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^3 z}{\partial y^3} + \frac{\partial z}{\partial y} + z = x + y$$

is a non-homogeneous PDE with constant coefficients.

### Reducible and Irreducible Linear Differential Operator

A linear differential operator  $F(D, D')$  is said to be reducible, if it can be written as a product of linear factors of the form  $(aD + bD' + c)$  with  $a, b, c$  as constants. For example, the operator  $F(D, D') = D^2 - D'^2$  is reducible because it can be written as

$$D^2 - D'^2 = (D - D')(D + D').$$

On the other hand, a differential operator  $F(D, D')$  is said to be irreducible if it cannot be written as a product of linear factors. For example,  $F(D, D') = D^2 - D'^3$  is irreducible.

### Reducible and Irreducible Linear PDE With Constant Coefficients

A linear PDE with constant coefficients  $F(D, D')z = f(x, y)$  is said to be reducible if  $F(D, D')$  is reducible and it is said to be irreducible if  $F(D, D')$  is irreducible. For example,  $F(D^2 - D'^2)z = x^2 y^3$  is reducible linear PDE with constant coefficients and whereas  $(D^2 - D'^3)z = x^2 y^3$  is irreducible.

## To find C.F. of a Reducible Non-homogeneous Linear PDE With Constant Coefficients

$$F(D, D')z = 0 \quad \dots \quad (1)$$

Let

$$F(D, D') = (b_1D - a_1D' - c_1)(b_2D - a_2D' - c_2) \dots (b_nD - a_nD' - c_n) \quad \dots \quad (2)$$

where  $a's, b's$  and  $c's$  are constants.

Then equation (1) reduces to

$$(b_1D - a_1D' - c_1)(b_2D - a_2D' - c_2) \dots (b_nD - a_nD' - c_n)z = 0 \quad \dots \quad (3)$$

Equation (3) shows that any solution of the equation

$$(b_rD - a_rD' - c_r)z = 0 \quad \dots \quad (4)$$

$r = 1, 2, \dots, n$  is a solution of (3).

(4) implies

$$b_r p - a_r q = c_r z$$

which is of the form

$$Pp + Qq = R.$$

It's Lagrange's A.E. are

$$\frac{dx}{b_r} = \frac{dy}{-a_r} = \frac{dz}{zc_r}$$

First two ratios imply

$$\begin{aligned} a_r dx + b_r dy &= 0 \\ \Rightarrow a_r x + b_r y &= c_1 \end{aligned}$$

Next, if  $b_r \neq 0$  from first and last ratios, we get

$$\begin{aligned} \frac{dx}{b_r} &= \frac{dz}{zc_r} \\ \Rightarrow \frac{c_r}{b_r} dx &= \frac{dz}{z} \\ \Rightarrow \frac{c_r}{b_r} x + \ln c_2 &= \ln z \\ \Rightarrow \frac{c_r}{b_r} x &= \ln \frac{z}{c_2} \\ \Rightarrow z &= c_2 e^{\frac{c_r x}{b_r}}, \text{ if } b_r \neq 0. \end{aligned}$$

Similarly, if  $a_r \neq 0$

$$z = c_2 e^{-\frac{c_r y}{a_r}}.$$

Therefore, the general solution of (4) is

$$\begin{aligned} c_2 &= \phi(c_1) \\ \Rightarrow z e^{-\frac{c_r x}{b_r}} &= \phi(a_r x + b_r y) \\ \Rightarrow z &= e^{\frac{c_r x}{b_r}} \phi(a_r x + b_r y), \quad \text{if } b_r \neq 0. \end{aligned}$$

And

$z = e^{-\frac{c_r y}{a_r}} \psi(a_r x + b_r y)$ , if  $a_r \neq 0$ , where  $\phi$  and  $\psi$  are arbitrary functions.

The general solution of (3) and consequently that of (1) is the sum of the solutions of the equation of the form (4) corresponding to each factor in (2)

$$C.F. = e^{-\frac{c_1 y}{a_1}} \phi_1(b_1 y + a_1 x) + e^{-\frac{c_2 y}{a_2}} \phi_2(b_2 y + a_2 x) + \dots + e^{-\frac{c_n y}{a_n}} \phi_n(b_n y + a_n x)$$

if  $a_r \neq 0, r = 1, 2, \dots, n$

and

$$C.F. = e^{\frac{c_1}{b_1}y} \phi_1(b_1y + a_1x) + e^{\frac{c_2}{b_2}y} \phi_2(b_2y + a_2x) + \dots + e^{\frac{c_n}{b_n}y} \phi_n(b_ny + a_nx)$$

if  $b_r \neq 0, r = 1, 2, \dots, n$

### Case of Repeated Factors

If a factor  $(bD - aD' - C)$  is repeated  $r$  times, then

$$C.F. = e^{\frac{c}{b}x} \sum_{i=1}^r x^{i-1} \phi_i(by + ax), \text{ if } b \neq 0,$$

And

$$C.F. = e^{-\frac{c}{a}y} \sum_{i=1}^r x^{i-1} \phi_i(by + ax), \text{ if } a \neq 0.$$

### Working Rule for Finding C.F. of Reducible Non-homogeneous Linear PDE With Constant Coefficients

Let the given PDE be

$$F(D, D')z = f(x, y).$$

Factorize  $F(D, D')$  into linear factors. Then, use the following rules.

#### Rule I

Corresponding to each linear factor  $(bD - aD' - C)$ , the part of C.F. is taken as  $e^{\frac{c}{b}x} \phi(by + ax)$ , if  $b \neq 0$ .

#### Particular Cases

##### 1. If $C = 0$ in Rule I

Then, corresponding to each linear factor  $(bD - aD')$  the part of C.F. is  $\phi(by + ax)$ ,  $b \neq 0$ .

##### 2. If $a = 0$ in Rule I

Then, corresponding to each linear factors  $(bD - C)$  the part of C.F. is  $e^{\frac{c}{b}x} \phi(by)$ , if  $b \neq 0$ .

##### 3. If $a = c = 0$ and $b = 1$ in Rule I

Then, corresponding to each factor  $D$ , the part of C.F. is  $\phi(y)$

#### Rule II

Corresponding to a repeated factor  $(bD - aD' - C)^r$ , the part of C.F. is

$$e^{\frac{c}{b}x} [\phi_1(by + ax) + x\phi_2(by + ax) + \dots + x^{r-1}\phi_r(by + ax)], b \neq 0.$$

#### Particular Cases

##### 1. If $C = 0$ in Rule II

Then, corresponding to each repeated factor  $(bD - aD')^r$ , the part of C.F. is

$$\phi_1(by + ax) + x\phi_2(by + ax) + \dots + x^{r-1}\phi_r(by + ax), b \neq 0.$$

##### 2. If $a = 0$ in Rule II

Then, corresponding to each repeated factor  $(bD - C)^r$ , the part of C.F. is

$$e^{\frac{c}{b}x} [\phi_1(by) + x\phi_2(by) + \dots + x^{r-1}\phi_r(by)], b \neq 0.$$

##### 3. If $a = c = 0$ and $b = 1$ in Rule II

Then, corresponding to repeated factor  $D^r$ , the part of C.F. is

$$\phi_1(y) + x\phi_2(y) + \dots + x^{r-1}\phi_r(y)$$

#### Rule III

Corresponding to each non-repeated linear factor  $(bD - aD' - c)$ , the part of C.F. is

$$e^{-\frac{c}{a}y} \phi(by + ax), \text{ if } a \neq 0.$$

#### Particular Cases

**1. If  $c = 0$  in Rule III**

Then, corresponding to each linear factor  $(bD - aD')$ , the part of C.F. is  $\phi(by + ax)$ ,  $a \neq 0$ .

**2. If  $b = 0$  in Rule III**

Then, corresponding to each linear factor  $(aD' - c)$ , the part of C.F. is  $e^{-\frac{c}{a}y} \phi(ax)$ ,  $a \neq 0$ .

**3. If  $b = c = 0$  and  $a = 1$  in Rule III**

Then, corresponding to each linear factor  $D'$ , the part of C.F. is  $\phi(x)$ .

**Rule IV**

Corresponding to a repeated factor  $(bD - aD' - c)^r$ , the part of C.F. is

$$e^{-\frac{c}{a}y} [\phi_1(by + ax) + y\phi_2(by + ax) + \dots + y^{r-1}\phi_r(by + ax)], a \neq 0.$$

**Particular Cases****1. If  $c = 0$  in Rule IV**

Then, corresponding to repeated factor  $(bD - aD')^r$ , the part of C.F. is

$$\phi_1(by + ax) + y\phi_2(by + ax) + \dots + y^{r-1}\phi_r(by + ax), a \neq 0.$$

**2. If  $b = 0$  in Rule IV**

Then, corresponding to repeated factor  $(aD' - c)^r$ , the part of C.F. is

$$e^{-\frac{c}{a}y} [\phi_1(ax) + y\phi_2(ax) + \dots + y^{r-1}\phi_r(ax)], a \neq 0.$$

**3. If  $b = c = 0, a = 1$ , in Rule IV**

Then, corresponding to repeated factor  $D'^r$ , the part of C.F. is

$$\phi_1(x) + y\phi_2(x) + \dots + y^{r-1}\phi_r(x).$$

**Examples**

1. Solve  $(D^2 - D'^2 + D - D')z = 0$

**Solution**

The given PDE is

$$(D^2 - D'^2 + D - D')z = 0 \quad \dots \quad (1)$$

$$\Rightarrow (D - D')(D + D' + 1)z = 0$$

$\therefore C.F. = \phi_1(y + x) + e^{-x}\phi_2(y - x)$ , where  $\phi_1$  and  $\phi_2$  are arbitrary functions.

Since  $f(x, y) = 0, P.I. = 0$ .

Therefore, the general solution is

$$z = \phi_1(y + x) + e^{-x}\phi_2(y - x)$$

2. Solve  $(D + D' - 1)(D + 2D' - 2)z = 0$

**Solution**

the Given PDE is

$$(D + D' - 1)(D + 2D' - 2)z = 0 \quad \dots \quad (1)$$

$\therefore C.F. = e^x\phi_1(y - x) + e^{2x}\phi_2(y - 2x)$  where  $\phi_1$  and  $\phi_2$  are arbitrary functions.

Since  $f(x, y) = 0, P.I. = 0$ .

Hence, the general solution is

$$z = e^x\phi_1(y - x) + e^{2x}\phi_2(y - 2x)$$

3. Solve  $(D^2 - a^2D'^2 + 2abD - 2a^2bD')z = 0$ .

**Solution**

The given equation is

$$(D^2 - a^2D'^2 + 2abD - 2a^2bD')z = 0$$

$$\text{or, } (D - aD')(D + aD' + 2ab)z = 0$$

$$\therefore C.F. = \phi_1(y + ax) + e^{-2abx}\phi_2(y - ax)$$

Since  $f(x, y) = 0, P.I. = 0$



Hence, the general solution is

$z = \phi_1(y + ax) + e^{-2abx}\phi_2(y - ax)$ , where  $\phi_1$  and  $\phi_2$  are arbitrary functions.

4. Solve  $r + 2s + t + 2p + 2q + z = 0$

**Solution**

The given equation is

$$\begin{aligned} r + 2s + t + 2p + 2q + z &= 0. \\ \text{or, } (D^2 + 2DD' + D'^2 + 2D + 2D' + 1)z &= 0 \\ \text{or, } (D + D' + 1)^2 z &= 0 \end{aligned}$$

This is the case of repeated linear factors.

$$\therefore C.F. = e^{-x}[\phi_1(y - x) + x(y - x)]$$

Since  $f(x, y) = 0, P.I. = 0$ .

Hence, the general solution is

$z = e^{-x}[\phi_1(y - x) + x(y - x)]$ , where  $\phi_1$  and  $\phi_2$  are arbitrary functions.

5. Solve  $DD'(3D - 5)(7D' + 2)(2D + 3D' + 5)z = 0$

**Solution**

The given PDE is

$$DD'(3D - 5)(7D' + 2)(2D + 3D' + 5)z = 0.$$

The C.F. is

$$C.F. = \phi_1(y) + \phi_2(x) + e^{\frac{5}{3}x}\phi_3(3y) + e^{-\frac{2}{7}y}\phi_4(7x) + e^{-\frac{5}{2}x}\phi_5(2y - 3x).$$

Since  $f(x, y) = 0, P.I. = 0$ .

Hence, the general solution is

$$z = \phi_1(y) + \phi_2(x) + e^{\frac{5}{3}x}\phi_3(3y) + e^{-\frac{2}{7}y}\phi_4(7x) + e^{-\frac{5}{2}x}\phi_5(2y - 3x).$$

where  $\phi_1, \phi_2, \phi_3, \phi_4, \phi_5$  are arbitrary functions.

### Method of Finding C.F. of Irreducible Linear PDE With Constant Coefficients

The C.F. of irreducible equation

$$F(D, D')z = f(x, y) \text{ is as } C.F. = \sum Ae^{hx+ky}$$

where  $A, h, k$  are arbitrary constants such that  $F(h, k) = 0$

### Examples

1. Solve  $(D - D'^2)z = 0$

**Solution**

Here the given equation is

$$(D - D'^2)z = 0 \quad \dots \quad (1)$$

Here  $F(D, D') = D - D'^2$ , which is irreducible.

Let

$$z = Ae^{hx+ky}$$

be a trial solution of (1). Then,

$$Dz = Ahe^{hx+ky}$$

And

$$D'^2 z = Ak^2 e^{hx+ky}$$

Putting these values in (1), we get

$$A(h - k^2)e^{hx+ky} = 0$$

And this implies

$$h - k^2 = 0 \Rightarrow h = k^2.$$

Replacing  $h$  by  $k^2$ , the general solution of (1) is

$$z = \sum A e^{k^2 x + ky}$$

where  $A$  and  $k$  are arbitrary constants.

$$2. \text{ Solve } (D - 2D' - 1)(D - 2D'^2 - 1)z = 0$$

**Solution**

The given equation is

$$(D - 2D' - 1)(D - 2D'^2 - 1)z = 0 \quad \dots \quad (1)$$

The C.F. corresponding to the linear factor  $(D - 2D' - 1)$  is

$$e^x \phi(y + 2x),$$

where  $\phi$  is an arbitrary function.

Next, let the trial solution corresponding to the non-linear factor  $(D - 2D'^2 - 1)$  be

$$z = A e^{hx + ky}$$

$$\therefore Dz = A h e^{hx + ky}$$

And

$$D'^2 z = A k^2 e^{hx + ky}.$$

So, putting these values in

$$(D - 2D'^2 - 1)z = 0$$

we get

$$(D - 2D'^2 - 1)e^{hx + ky} = 0$$

$$\Rightarrow (h - 2k^2 - 1) = 0$$

$$\Rightarrow h = 2k^2 + 1$$

Replacing  $h$  by  $2k^2 + 1$ , the C.F. corresponding to the non-linear factor  $D - 2D'^2 - 1$  is

$$\sum A e^{(2k^2 + 1)x + ky}$$

where  $A$  and  $k$  are arbitrary constants.

Hence, the required solution is

$$z = e^x \phi(y + 2x) + \sum A e^{(2k^2 + 1)x + ky}$$

$$3. \text{ Solve } (2D^4 - 3D^2 D' + D'^2)z = 0$$

**Solution**

The given PDE is

$$(2D^4 - 3D^2 D' + D'^2)z = 0$$

$$\text{or, } (2D^2 - D')(D^2 - D')z = 0 \quad \dots \quad (1)$$

Let

$$z = A e^{hx + ky}$$

be a trial solution of

$$(2D^2 - D')z = 0$$

$$\therefore (2h^2 - k)A e^{hx + ky} = 0$$

$$\Rightarrow 2h^2 - k = 0$$

$$\Rightarrow k = 2h^2$$

$\therefore$  C.F. corresponding to the factor  $2D^2 - D'$  is

$$\sum A e^{hx + 2h^2 y} \quad \dots \quad (2)$$

Again let

$$z = A'e^{h'x+k'y}$$

be a trial solution of

$$\begin{aligned}(D^2 - D')z &= 0 \\ \therefore (h'^2 - k')A'e^{h'x+k'y} &= 0 \\ \Rightarrow h'^2 - k' &= 0 \\ \Rightarrow k' &= h'^2\end{aligned}$$

$\therefore$  C.F. corresponding to  $(D^2 - D')z = 0$  is

$$\sum A'e^{h'x+h'^2y} \quad \dots \quad (3)$$

Hence, the general solution of (1) is

$$z = \sum Ae^{hx+2h^2y} + \sum A'e^{h'x+h'^2y}$$

where  $A, A', h, h', k, k'$  are arbitrary constants.

4. Solve  $(D^2 - 4DD' + D - 1)z = e^{3x-2y}$

**Solution**

The given PDE is

$$(D^2 - 4DD' + D - 1)z = e^{3x-2y} \quad \dots \quad (1)$$

Here  $F(D, D') = D^2 - 4DD' + D - 1$  is irreducible.

So, let a trial solution of

$$\begin{aligned}(D^2 - 4DD' + D - 1)z &= 0 \quad \dots \quad (2), be \\ z &= Ae^{hx+ky}\end{aligned}$$

Therefore,

$$\begin{aligned}D^2z &= Ah^2e^{hx+ky} \\ DD'z &= Ahke^{hx+ky} \\ Dz &= Ahe^{hx+ky}\end{aligned}$$

Thus, (2) reduces to

$$\begin{aligned}A(h^2 - 4hk + h - 1)e^{hx+ky} &= 0 \\ \Rightarrow h^2 - 4hk + h - 1 &= 0 \\ \Rightarrow k &= \frac{h^2 + h - 1}{4h} \quad \dots \quad (3) \\ \therefore C.F. &= \sum Ae^{hx+ky},\end{aligned}$$

where  $k$  is given by (3).

Again

$$\begin{aligned}P.I. &= \frac{1}{F(D, D')}e^{3x-2y} \\ &= \frac{1}{D^2 - 4DD' + D - 1}e^{3x-2y} \\ &= \frac{1}{3^2 - 4 \cdot 3 \cdot (-2) + 3 - 1} \int \int e^v dv dv, \text{ where } v = 3x - 2y \\ &= \frac{1}{35}e^v = \frac{1}{35}e^{3x-2y}.\end{aligned}$$

Hence, the required general solution is

$$z = C.F. + P.I.$$

That is,

$$z = \sum A e^{hx+ky} + \frac{1}{35} e^{3x-2y}$$

where  $A, h, k$  are arbitrary constant where  $h$  and  $k$  are related by (3).

### To Find P.I. of Non-homogeneous Linear PDE (Reducible or Irreducible)

Suppose  $F(D, D')z = f(x, y)$  ... (1)

be non-homogeneous linear PDE (reducible or irreducible)

#### Case I

When  $f(x, y) = e^{ax+by}$  and  $F(a, b) \neq 0$  then,

$$P.I. = \frac{1}{F(D, D')} e^{ax+by} = \frac{1}{F(a, b)} e^{ax+by}$$

#### Case II

When  $f(x, y) = \sin(ax + by)$  or  $\cos(ax + by)$ , then

$$P.I. = \frac{1}{F(D, D')} \sin(ax + by)$$

Or

$$P.I. = \frac{1}{F(D, D')} \cos(ax + by)$$

which is evaluated by putting  $D^2 = -a^2, D'^2 = -b^2, DD' = -ab$ , provided the denominator is non-zero.

#### Case III

When  $f(x, y) = x^m y^n$ , then

$$P.I. = \frac{1}{F(D, D')} x^m y^n = [F(D, D')]^{-1} x^m y^n$$

#### Case IV

When  $f(x, y) = V e^{ax+by}$ , where  $V$  is a function of  $x$  and  $y$ , then

$$P.I. = \frac{1}{F(D, D')} V e^{ax+by} = e^{ax+by} \frac{1}{F(D + a, D' + b)} V.$$

#### Note

If  $F(a, b) = 0$  and  $f(x, y) = e^{ax+by}$ , then

$$P.I. = \frac{1}{F(D, D')} e^{ax+by} = e^{ax+by} \frac{1}{F(D + a, D' + b)} \cdot 1$$

that is, taking  $V = 1$  in case IV.

#### Examples

1. Solve  $(DD' + aD + bD' + ab)z = e^{mx+ny}$

Solution

The given PDE is

$$(DD' + aD + bD' + ab)z = e^{mx+ny} \quad \dots \quad (1)$$

$$\text{or, } (D + b)(D' + a)z = e^{mx+ny}$$

$$\therefore C.F. = e^{-bx} \phi_1(y) + e^{ay} \phi_2(x)$$

where  $\phi_1, \phi_2$  are arbitrary functions.

#### To find P.I.

$$\begin{aligned} P.I. &= \frac{1}{F(D, D')} e^{mx+ny} \\ &= \frac{1}{(D + b)(D' + a)} e^{mx+ny} \end{aligned}$$

$$= \frac{1}{(m+b)(n+a)} e^{mx+xy}$$

Hence, the general solution is

$$z = C.F. + P.I.$$

$$z = e^{-bx} \phi_1(y) + e^{ay} \phi_2(x) + \frac{1}{(m+b)(n+a)} e^{mx+xy}$$

2. Solve  $(D^2 - 4DD' + D - 1)z = e^{3x-2y}$ .

**Solution**

The given PDE is

$$(D^2 - 4DD' + D - 1)z = e^{3x-2y} \quad \dots \quad (1)$$

Here  $F(D, D') = D^2 - 4DD' + D - 1$  is irreducible. So, for C.F., consider

$$(D^2 - 4DD' + D - 1)z = 0 \quad \dots \quad (2)$$

Let  $z = Ae^{hx+ky}$  be a trial solution of (2)

$$\therefore D^2 z = Ah^2 e^{hx+ky}, DD' z = Ahke^{hx+ky}, Dz = Ahe^{hx+ky}$$

Therefore, (1) reduces to

$$\begin{aligned} A(h^2 - 4hk + h - 1)e^{hx+ky} &= 0 \\ \Rightarrow h^2 - 4hk + h - 1 &= 0 \\ \Rightarrow k &= \frac{h^2 + h - 1}{4h} \quad \dots \quad (3) \end{aligned}$$

$\therefore C.F. = \sum Ae^{hx+ky}$ , where  $k$  is given by (3).

**To find P.I.**

$$\begin{aligned} P.I. &= \frac{1}{F(D, D')} e^{3x-2y} \\ &= \frac{1}{D^2 - 4DD' + D - 1} e^{3x-2y} \\ &= \frac{1}{3^2 - 4 \times 3 \times (-2) + 3 - 1} e^{3x-2y} \\ &= \frac{1}{35} e^{3x-2y} \end{aligned}$$

Hence, the required general solution is

$$\begin{aligned} z &= C.F. + P.I. \\ z &= \sum Ae^{hx+ky} + \frac{1}{35} e^{3x-2y}, \end{aligned}$$

where  $A, h, k$  are arbitrary constant and  $h$  and  $k$  are related by (3).

3. Solve  $(D^2 + DD' + D' - 1)z = \sin(x + 2y)$

**Solution**

The given equation is

$$\begin{aligned} (D^2 + DD' + D' - 1)z &= \sin(x + 2y) \quad \dots \quad (1) \\ (D + 1)(D + D' - 1)z &= \sin(x + 2y) \\ \therefore C.F. &= e^{-x} \phi_1(y) + e^x \phi_2(y - x), \end{aligned}$$

where  $\phi_1$ , and  $\phi_2$  are arbitrary functions.

**To find P.I.**

$$\begin{aligned} P.I. &= \frac{1}{F(D, D')} \sin(x + 2y) \\ &= \frac{1}{D^2 + DD' + D' - 1} \sin(x + 2y) \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{-1^2 - (1.2) + D' - 1} \sin(x + 2y) \\
&= \frac{1}{D' - 4} \sin(x + 2y) \\
&= (D' + 4) \frac{1}{D'^2 - 16} \sin(x + 2y) \\
&= (D' + 4) \frac{1}{-2^2 - 16} \sin(x + 2y) \\
&= (D' + 4) \times \frac{1}{-20} \sin(x + 2y) \\
&= \frac{1}{-20} [(D' + 4) \sin(x + 2y)] \\
&= \frac{1}{-20} [2 \cos(x + 2y) + 4 \sin(x + 2y)]
\end{aligned}$$

Hence, the general solution is

$$z = C.F. + P.I.$$

$$z = e^{-x} \phi_1(y) + e^x \phi_2(y - x) - \frac{1}{10} [\cos(x + 2y) + 4 \sin(x + 2y)]$$

4. Solve  $r - s + 2q - z = x^2 y^2$

**Solution**

The given PDE is

$$\begin{aligned}
r - s + 2q - z &= x^2 y^2 \quad \dots (1) \\
\Rightarrow (D^2 - DD' + 2D' - 1)z &= x^2 y^2
\end{aligned}$$

Since,  $D^2 - DD' + 2D' - 1$  is irreducible, then for C.F. we consider

$$(D^2 - DD' + 2D' - 1)z = 0 \quad \dots (2)$$

Let  $z = Ae^{hx+ky}$  be a trial solution of (2)

$$\therefore D^2 z = Ah^2 e^{hx+ky}, DD' z = Ahke^{hx+ky}, D' z = Ake^{hx+ky}$$

Then, (2) reduces to

$$\begin{aligned}
A(h^2 - hk + 2k - 1)e^{hx+ky} &= 0 \\
\text{or, } h^2 - hk + 2k - 1 &= 0 \\
\therefore k &= \frac{1 - h^2}{2 - h} \quad \dots (3) \\
\therefore C.F. &= \sum Ae^{hx+ky}
\end{aligned}$$

where  $A, h, k$  are arbitrary constants,  $h$  and  $k$  are related by (3).

**To find P.I.**

$$\begin{aligned}
P.I. &= \frac{1}{F(D, D')} x^2 y^2 \\
&= \frac{1}{D^2 - DD' + 2D' - 1} x^2 y^2 \\
&= \frac{-1}{1 - (D^2 - DD' + 2D')} x^2 y^2 \\
&= -[1 - (D^2 - DD' + 2D')]^{-1} x^2 y^2 \\
&= -[1 + (D^2 - DD' + 2D') + (D^2 - DD' + 2D')^2 + (D^2 - DD' + 2D')^3 + \dots] x^2 y^2 \\
&= -[1 + D^2 - DD' + 2D' + D^2 D'^2 + 4D'^2 + 4D^2 D' - 4DD'^2 + 12D^2 D'^2 + \dots] x^2 y^2 \\
&= -[x^2 y^2 + 2y^2 - 4xy + 4x^2 y + 8x^2 + 16x + 16y + 52] \\
&= -x^2 y^2 - 2y^2 + 4xy - 4x^2 y - 8x^2 - 16x - 16y - 52
\end{aligned}$$

Therefore, the required general solution is

$$z = C.F. + P.I.$$

$$\therefore z = \sum Ae^{hx+ky} - x^2y^2 - 2y^2 + 4xy - 4x^2y - 8x^2 - 16x - 16y - 52$$

5. Solve  $(3D^2 - 2D'^2 + D - 1)z = 4e^{x+y}\cos(x+y)$

**Solution**

The given PDE is

$$(3D^2 - 2D'^2 + D - 1)z = 4e^{x+y}\cos(x+y) \quad \dots \quad (1)$$

Since  $F(D, D') = 3D^2 - 2D'^2 + D - 1$  is irreducible to find C.F. we consider the equation.

$$(3D^2 - 2D'^2 + D - 1)z = 0 \quad \dots \quad (2)$$

Let  $z = Ae^{hx+ky}$  be a trial solution of (2). Then,

$$D^2z = Ah^2e^{hx+ky}, D'^2z = Ak^2e^{hx+ky}, Dz = Ahe^{hx+ky}$$

$\therefore$  (2) reduces to

$$A(3h^2 - 2k^2 + h - 1)e^{hx+ky} = 0 \\ \Rightarrow 3h^2 - 2k^2 + h - 1 = 0 \quad \dots \quad (3)$$

$$\therefore C.F. = \sum Ae^{hx+ky},$$

where  $A, h, k$  are constants,  $h$  and  $k$  are related by (3).

**To find P.I.**

$$\begin{aligned} P.I. &= \frac{1}{F(D, D')} 4e^{x+y}\cos(x+y) \\ &= 4 \frac{1}{3D^2 - 2D'^2 + D - 1} e^{x+y}\cos(x+y) \\ &= 4e^{x+y} \frac{1}{3(D+1)^2 - 2(D'+1)^2 + (D+1) - 1} \cos(x+y) \\ &= 4e^{x+y} \frac{1}{3D^2 + 7D - 2D'^2 - 4D' + 1} \cos(x+y) \\ &= 4e^{x+y} \frac{1}{3(-1^2) + 7D - 2(-1^2) - 4D' + 1} \cos(x+y) \\ &= 4e^{x+y} \frac{1}{7D - 4D'} \cos(x+y) \\ &= 4e^{x+y} \frac{7D + 4D'}{49D^2 - 16D'^2} \cos(x+y) \\ &= 4e^{x+y} \frac{7D + 4D'}{49(-1^2) - 16(-1^2)} \cos(x+y) \\ &= 4e^{x+y} \frac{7D + 4D'}{-33} \cos(x+y) \\ &= \frac{4e^{x+y}}{-33} (7D\cos(x+y) + 4D'\cos(x+y)) \\ &= \frac{4e^{x+y}}{-33} (-7\sin(x+y) - 4\sin(x+y)) \\ &= \frac{4e^{x+y}}{-33} (-11\sin(x+y)) \\ &= \frac{4e^{x+y}\sin(x+y)}{3} \end{aligned}$$

Hence, the general solution is

$$z = C.F. + P.I.$$

That is,

$$z = \sum A e^{hx+ky} + \frac{4e^{x+y} \sin(x+y)}{3}$$

where  $A, h, k$  are constants and  $h, k$  are related by (3).

### Second Order Linear PDE of Order Two With Variable Coefficients.

The most general linear PDE of second order in two independent variables  $x$  and  $y$  with variable coefficients is of the form  $Rr + Ss + Tt + Pp + Qq + Zz = F$ , where  $R, S, T, P, Q, Z$  and  $F$  are functions of  $x$  and  $y$  only and not all  $R, S, T$  are zero.

### Classification of PDE of Second Order

Consider a general PDE of second order for a function of two independent variables  $x$  and  $y$  of the form

$$Rr + Ss + Tt + f(x, y, z, p, q) = 0 \quad \dots \quad (1)$$

where  $R, S, T$  are continuous functions of  $x$  and  $y$  only and having partial derivatives defined in some domain  $D$  on  $xy$  -plane. Then, (1) is said to be

- i. Hyperbolic at a point  $(x, y)$  in  $D$  if  $S^2 - 4RT > 0$
- ii. Parabolic at a point  $(x, y)$  in  $D$  if  $S^2 - 4RT = 0$
- iii. Elliptic at a point  $(x, y)$  in the domain  $D$  if  $S^2 - 4RT < 0$ .

### Examples

1. Classify  $2 \left( \frac{\partial^2 u}{\partial x^2} \right) + 4 \left( \frac{\partial^2 u}{\partial x \partial y} \right) + 3 \left( \frac{\partial^2 u}{\partial y^2} \right) = 2$

### Solution

The given PDE is

$$2 \left( \frac{\partial^2 u}{\partial x^2} \right) + 4 \left( \frac{\partial^2 u}{\partial x \partial y} \right) + 3 \left( \frac{\partial^2 u}{\partial y^2} \right) = 2$$

or,  $2r + 4s + 3t = 2 \quad \dots \quad (1)$

Comparing it with

$Rr + Ss + Tt + f(x, y, z, p, q) = 0$ , we get

$$R = 2, S = 4, T = 3.$$

Now,

$$S^2 - 4RT$$

$$4^2 - 4.2.3 = -8 < 0.$$

Therefore, the equation is elliptic at every point in its domain.

2.  $\frac{\partial^2 u}{\partial x^2} + 4 \frac{\partial^2 u}{\partial x \partial y} + 4 \frac{\partial^2 u}{\partial y^2} = 0$

### Solution

The given PDE is

$$\frac{\partial^2 u}{\partial x^2} + 4 \frac{\partial^2 u}{\partial x \partial y} + 4 \frac{\partial^2 u}{\partial y^2} = 0$$

or,  $r + 4s + 4t = 0 \quad \dots \quad (1)$

Comparing it with

$$Rr + Ss + Tt + f(x, y, z, p, q) = 0$$

we get

$$R = 1, S = 4, T = 4$$

Now,

$$S^2 - 4RT$$

$$= 16 - 4.1.4 = 0$$



Therefore, the given equation is parabolic at every point of it's domain.

iii.  $xyr - (x^2 - y^2)s - xyt + py - qx = 2(x^2 - y^2)$

**Solution**

$$xyr - (x^2 - y^2)s - xyt + py - qx = 2(x^2 - y^2) \quad \dots (1)$$

Comparing it with

$$Rr + Ss + Tt + f(x, y, z, p, q) = 0$$

we have

$$R = xy, S = -(x^2 - y^2), T = -xy.$$

So,

$$\begin{aligned} & S^2 - 4RT \\ &= \{-(x^2 - y^2)\}^2 - 4(xy) \times (-xy) \\ &= (x^2 - y^2)^2 + 4x^2y^2 \\ &= (x^2 + y^2)^2 > 0 \text{ for all values of } x \text{ and } y. \end{aligned}$$

So, the equation is hyperbolic at every point of it's domain.

### Laplace Transformation

#### Reduction to Canonical (or Normal) Forms

Consider a PDE of the type

$$Rr + Ss + Tt + f(x, y, z, p, q) = 0 \quad \dots (1)$$

where  $R, S, T$  are continuous functions of  $x$  and  $y$  having continuous partial derivatives of every order. Laplace transformation of (1) is to change the independent variables  $x$  and  $y$  to a new set of continuously differentiable independent variables  $u$  and  $v$  where,

$$u = u(x, y) \text{ and } v = v(x, y) \quad \dots (2)$$

are to be chosen that the resulting equation in  $u$  and  $v$  is transformed into one of three canonical forms, which are easily integrable.

$$\begin{aligned} p &= \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} \\ q &= \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} \end{aligned} \quad \dots (3)$$

(3) implies

$$\begin{aligned} \frac{\partial}{\partial x} &= \frac{\partial u}{\partial x} \cdot \frac{\partial}{\partial u} + \frac{\partial v}{\partial x} \cdot \frac{\partial}{\partial v} \\ \frac{\partial}{\partial y} &= \frac{\partial u}{\partial y} \cdot \frac{\partial}{\partial u} + \frac{\partial v}{\partial y} \cdot \frac{\partial}{\partial v} \end{aligned} \quad \dots (4)$$

$$\begin{aligned} \therefore r &= \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left( \frac{\partial z}{\partial x} \right) \\ &= \left( \frac{\partial u}{\partial x} \cdot \frac{\partial}{\partial u} + \frac{\partial v}{\partial x} \cdot \frac{\partial}{\partial v} \right) \left( \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} \right), \text{ using (3) and (4)} \\ &= \frac{\partial u}{\partial x} \cdot \frac{\partial}{\partial u} \left( \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} \right) + \frac{\partial v}{\partial x} \cdot \frac{\partial}{\partial v} \left( \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} \right) \\ &= \frac{\partial^2 z}{\partial u^2} \left( \frac{\partial u}{\partial x} \right)^2 + 2 \frac{\partial^2 z}{\partial u \partial v} \cdot \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial x} + \frac{\partial^2 z}{\partial v^2} \left( \frac{\partial v}{\partial x} \right)^2 + \frac{\partial z}{\partial u} \cdot \frac{\partial^2 u}{\partial x^2} + \frac{\partial z}{\partial v} \cdot \frac{\partial^2 v}{\partial x^2} \\ s &= \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left( \frac{\partial z}{\partial y} \right) \\ &= \left( \frac{\partial u}{\partial x} \cdot \frac{\partial}{\partial u} + \frac{\partial v}{\partial x} \cdot \frac{\partial}{\partial v} \right) \left( \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} \right), \text{ by (3) and (4)} \end{aligned}$$

$$= \frac{\partial^2 z}{\partial u^2} \cdot \frac{\partial u}{\partial x} \cdot \frac{\partial u}{\partial y} + \frac{\partial^2 z}{\partial u \partial v} \left( \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial y} + \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial x} \right) + \frac{\partial^2 z}{\partial v^2} \cdot \frac{\partial v}{\partial x} \cdot \frac{\partial v}{\partial y} + \frac{\partial z}{\partial u} \cdot \frac{\partial^2 u}{\partial y \partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial^2 v}{\partial y \partial x}$$

and

$$\begin{aligned} t &= \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left( \frac{\partial z}{\partial y} \right) \\ &= \left( \frac{\partial u}{\partial y} \cdot \frac{\partial}{\partial u} + \frac{\partial v}{\partial y} \cdot \frac{\partial}{\partial v} \right) \left( \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} \right), \quad \text{using (3) and (4)} \\ &= \frac{\partial u}{\partial y} \cdot \frac{\partial}{\partial u} \left( \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} \right) + \frac{\partial v}{\partial y} \cdot \frac{\partial}{\partial v} \left( \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} \right) \\ &= \frac{\partial^2 z}{\partial u^2} \cdot \left( \frac{\partial u}{\partial y} \right)^2 + 2 \frac{\partial^2 z}{\partial u \partial v} \cdot \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial y} + \frac{\partial^2 z}{\partial v^2} \cdot \left( \frac{\partial v}{\partial y} \right)^2 + \frac{\partial z}{\partial u} \cdot \frac{\partial^2 u}{\partial y^2} + \frac{\partial z}{\partial v} \cdot \frac{\partial^2 v}{\partial y^2} \end{aligned}$$

Putting these values of  $p, q, r, s, t$  in (1) and simplifying, we get

$$A \frac{\partial^2 z}{\partial u^2} + 2B \frac{\partial^2 z}{\partial u \partial v} + C \frac{\partial^2 z}{\partial v^2} + F \left( u, v, z, \frac{\partial z}{\partial u}, \frac{\partial z}{\partial v} \right) = 0 \quad \dots (5)$$

Where,

$$A = R \left( \frac{\partial u}{\partial x} \right)^2 + S \frac{\partial u}{\partial x} \cdot \frac{\partial u}{\partial y} + T \left( \frac{\partial u}{\partial y} \right)^2 \quad \dots (6)$$

$$B = R \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial x} + \frac{1}{2} S \left( \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial y} + \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial x} \right) + T \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial y} \quad \dots (7)$$

$$C = R \left( \frac{\partial v}{\partial x} \right)^2 + S \frac{\partial v}{\partial x} \cdot \frac{\partial v}{\partial y} + T \left( \frac{\partial v}{\partial y} \right)^2 \quad \dots (8)$$

And

$$F \left( u, v, z, \frac{\partial z}{\partial u}, \frac{\partial z}{\partial v} \right)$$

is the transformed form of

$$f(x, y, z, p, q).$$

Now, we shall find out  $u$  and  $v$  so that (5) reduces to simplest possible form. The method of evaluation of desired values of  $u$  and  $v$  becomes easy when the discriminant  $S^2 - 4RT$  of the quadratic equation.

$$R\lambda^2 + S\lambda + T = 0 \quad \dots (9)$$

is everywhere either  $+ve$  or  $-ve$  or zero.

### Case I [When $S^2 - 4RT > 0$ ]

If  $S^2 - 4RT > 0$ , then the roots  $\lambda_1$  and  $\lambda_2$  of (9) are real and distinct.

The coefficient of  $\frac{\partial^2 z}{\partial u^2}$  and  $\frac{\partial^2 z}{\partial v^2}$  in (9) will be zero if we choose  $u$  and  $v$  such that

$$\frac{\partial u}{\partial x} = \lambda_1 \frac{\partial u}{\partial y} \quad \dots (10)$$

And

$$\frac{\partial v}{\partial x} = \lambda_2 \frac{\partial v}{\partial y} \quad \dots (11)$$

Since  $\lambda_1$  is a root of (9), we have

$$R\lambda_1^2 + S\lambda_1 + T = 0 \quad \dots (12)$$

Using (10) in (6) we get

$$A = (R\lambda_1^2 + S\lambda_1 + T) \left( \frac{\partial u}{\partial y} \right)^2 = 0 \quad \dots (13), \quad \text{by (12)}$$

Again since  $\lambda_2$  is a root of (9), we have

$$R\lambda_2^2 + S\lambda_2 + T = 0 \quad \dots \quad (14)$$

Using (11) in (9) we get

$$C = (R\lambda_2^2 + S\lambda_2 + T) \left( \frac{\partial v}{\partial y} \right)^2 = 0 \quad \dots \quad (15), \quad \text{by (14)}$$

From (15), we have

$$\frac{\partial u}{\partial x} - \lambda_1 \frac{\partial u}{\partial y} = 0 \quad \dots \quad (16)$$

Lagrange's A.E. of (16) are

$$\frac{dx}{1} = \frac{dy}{-\lambda_1} = \frac{du}{0} \quad \dots \quad (17)$$

From third fraction, we have

$$du = 0 \Rightarrow u = c_1 \quad \dots \quad (18)$$

where  $c_1$  is an arbitrary constant.

From 1<sup>st</sup> and 2<sup>nd</sup> ratios of (17), we have

$$\frac{dy}{dx} + \lambda_1 = 0 \quad \dots \quad (19)$$

Let the solution of (19) be

$$f_1(x, y) = c_2 \quad \dots \quad (20)$$

$c_2$  being an arbitrary constant.

From (18) and (20), the general solution of (16) or that of (10) is

$$u = f_1(x, y) \quad \dots \quad (21)$$

Similarly, the general solution of (11) can be taken as

$$v = f_2(x, y) \quad \dots \quad (22)$$

Here,  $f_1$  and  $f_2$  are arbitrary functions.

We can verify that

$$\begin{aligned} AC - B^2 &= \frac{1}{4} [4RT - S^2] \left( \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} \right)^2 \\ \text{or, } B^2 &= \frac{1}{4} [S^2 - 4RT] \left( \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} \right)^2, \quad \dots \quad (23) \\ &\quad [\text{since } A = C = 0] \end{aligned}$$

Let the Jacobian of  $u$  and  $v$  be not zero. That is, let

$$J = \frac{\partial(u, v)}{\partial(x, y)} = \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial x} \neq 0.$$

Since  $S^2 - 4RT > 0$ , (23) shows that  $B^2 > 0$ . Hence, dividing both sides of (5) by  $B^2$ , we get

$$\begin{aligned} \frac{\partial^2 z}{\partial u \partial v} &= \phi \left( u, v, z, \frac{\partial z}{\partial u}, \frac{\partial z}{\partial v} \right) \quad \dots \quad (24) \\ &\quad [\text{Since } A = C = 0] \end{aligned}$$

This is the canonical form of (1) in this case.

### Case II [When $S^2 - 4RT = 0$ ]

If  $S^2 - 4RT = 0$ , the roots  $\lambda_1$  and  $\lambda_2$  of (9) are real and equal. We now take  $u$  exactly as in case I and take  $v$  as a function of  $x$  and  $y$  which is independent of  $u$ . As in first case  $A = 0$ . Also since  $S^2 - 4RT = 0$ , (23) show tht  $B^2 = 0$  and so  $B = 0$ . But  $C \neq 0$ , otherwise  $v$  would be a function of  $u$ . Putting  $A = B = 0$  and dividing by  $C \neq 0$ , equation (5) transforms to

$$\frac{\partial^2 z}{\partial v^2} = \phi \left( u, v, z, \frac{\partial z}{\partial u}, \frac{\partial z}{\partial v} \right) \quad \dots \quad (25)$$

which is the Canonical form of (1) when  $S^2 - 4RT = 0$ .

### Case III [When $S^2 - 4RT < 0$ ]

If  $S^2 - 4RT < 0$ , the roots  $\lambda_1$  and  $\lambda_2$  of (9) are complex imaginary). Thus, this case is formally the same as case I. Therefore, proceeding as in case I, we find that (1) reduces to (24) but the variables  $u, v$ , are complex conjugates instead of being real. To obtain real Canonical form we make further transformation  $u = \alpha + i\beta$  and  $v = \alpha - i\beta$  so that

$$\alpha = \frac{u + v}{2}, \quad \beta = \frac{i(v - u)}{2} \quad \dots \quad (26)$$

Now,

$$\begin{aligned} \frac{\partial z}{\partial u} &= \frac{\partial z}{\partial \alpha} \cdot \frac{\partial \alpha}{\partial u} + \frac{\partial z}{\partial \beta} \cdot \frac{\partial \beta}{\partial u} \\ &= \frac{1}{2} \left( \frac{\partial z}{\partial \alpha} - i \frac{\partial z}{\partial \beta} \right) \quad \dots \quad (27) \\ &\quad [\text{using (26)}] \end{aligned}$$

And

$$\begin{aligned} \frac{\partial z}{\partial v} &= \frac{\partial z}{\partial \alpha} \cdot \frac{\partial \alpha}{\partial v} + \frac{\partial z}{\partial \beta} \cdot \frac{\partial \beta}{\partial v} \\ &= \frac{1}{2} \left( \frac{\partial z}{\partial \alpha} + i \frac{\partial z}{\partial \beta} \right) \quad \dots \quad (28) \\ &\quad [\text{using (26)}] \end{aligned}$$

$$\begin{aligned} \therefore \frac{\partial^2 z}{\partial u \partial v} &= \frac{\partial}{\partial u} \left( \frac{\partial z}{\partial v} \right) = \frac{1}{2} \left[ \frac{\partial}{\partial \alpha} - i \frac{\partial}{\partial \beta} \right] \times \frac{1}{2} \left[ \frac{\partial z}{\partial \alpha} + i \frac{\partial z}{\partial \beta} \right] \\ &\quad [\text{using (27) and (28)}] \\ &= \frac{1}{4} \left[ \frac{\partial}{\partial \alpha} \left( \frac{\partial z}{\partial \alpha} + i \frac{\partial z}{\partial \beta} \right) - i \frac{\partial}{\partial \beta} \left( \frac{\partial z}{\partial \alpha} + i \frac{\partial z}{\partial \beta} \right) \right] \\ &= \frac{1}{4} \left[ \frac{\partial^2 z}{\partial \alpha^2} + i \frac{\partial^2 z}{\partial \alpha \partial \beta} - i \frac{\partial^2 z}{\partial \beta \partial \alpha} + \frac{\partial^2 z}{\partial \beta^2} \right] \\ \therefore \frac{\partial^2 z}{\partial u \partial v} &= \frac{1}{4} \left[ \frac{\partial^2 z}{\partial \alpha^2} + \frac{\partial^2 z}{\partial \beta^2} \right] \quad \dots \quad (29) \end{aligned}$$

Putting  $u = \alpha + i\beta$  and  $v = \alpha - i\beta$  and using (27), (28) and (29) equation (24) reduces to

$$\frac{\partial^2 z}{\partial \alpha^2} + \frac{\partial^2 z}{\partial \beta^2} = \psi \left( \alpha, \beta, z, \frac{\partial z}{\partial \alpha}, \frac{\partial z}{\partial \beta} \right) \quad \dots \quad (30)$$

Which is the Canonical form of (1) in case  $S^2 - 4RT < 0$ .

### Working Rule for Reducing a Hyperbolic Equation to it's Canonical Form

1. Let the given equation

$$Rr + Ss + Tt + f(x, y, z, p, q) = 0 \quad \dots \quad (1)$$

be hyperbolic so that

$$S^2 - 4RT > 0.$$

2. Write  $\lambda$  –quadratic equation  $R\lambda^2 + S\lambda + T = 0 \quad \dots \quad (2)$

Let  $\lambda_1, \lambda_2$  be two distinct roots of (2)

3. Then, corresponding characteristics equations are

$$\frac{dy}{dx} + \lambda_1 = 0 \text{ and } \frac{dy}{dx} + \lambda_2 = 0$$

Solving these, we get  $f_1(x, y) = c_1$  and  $f_2(x, y) = c_2$

$$\left. \begin{aligned} f_1(x, y) &= c_1 \\ \text{and } f_2(x, y) &= c_2 \end{aligned} \right\} \quad \dots \quad (3)$$

4. We choose  $u$  and  $v$  such that

$$\left. \begin{array}{l} u = f_1(x, y) \\ \text{and } v = f_2(x, y) \end{array} \right\} \dots \quad (4)$$

5. Using relation (4) find  $p, q, r, s, t$  in terms of  $u$  and  $v$ .
6. Substituting the values of  $p, q, r, s, t$ , obtained in step (5) in (1) and simplifying we will get the following canonical form of (10)

$$\frac{\partial^2 z}{\partial u \partial v} = \phi \left( u, v, z, \frac{\partial z}{\partial u}, \frac{\partial z}{\partial v} \right).$$

### Examples

1. Write the canonical form of  $\frac{\partial^2 z}{\partial x^2} - c^2 \frac{\partial^2 z}{\partial y^2} = 0$

#### Solution

The given equation is

$$\frac{\partial^2 z}{\partial x^2} - c^2 \frac{\partial^2 z}{\partial y^2} = 0$$

or,  $r - c^2 t = 0 \quad \dots \quad (1)$

Comparing it with

$$Rr + Ss + Tt + f(x, y, z, p, q) = 0,$$

we get

$$R = 1, S = 0, T = -1$$

$$\therefore S^2 - 4RT = 0 - 4.1(-1) = 4 > 0$$

Therefore, (1) is hyperbolic.

The  $\lambda$  quadratic equation is

$$R\lambda^2 + S\lambda + T = 0$$

$$\Rightarrow \lambda^2 - 1 = 0$$

$$\therefore \lambda = 1, -1, \text{ i.e. } \lambda_1 = 1, \lambda_2 = -1$$

(real and distinct roots)

Then, the characteristics equation are

$$\Rightarrow \frac{dy}{dx} + \lambda_1 = 0 \text{ and } \frac{dy}{dx} - 1 = 0$$

Integration implies

$$y + x = c_1 \text{ and } y - x = c_2$$

Let us choose

$$u = y + x, v = y - x \quad \dots \quad (2)$$

$$\therefore p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} = \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \quad \dots \quad (3)$$

And

$$q = \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \quad \dots \quad (4)$$

From (3) and (4), we have

$$\left. \begin{array}{l} \frac{\partial}{\partial x} = \frac{\partial}{\partial u} - \frac{\partial}{\partial v} \\ \text{and } \frac{\partial}{\partial y} = \frac{\partial}{\partial u} + \frac{\partial}{\partial v} \end{array} \right\} \quad \dots \quad (5)$$

$$r = \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left( \frac{\partial z}{\partial x} \right) = \left( \frac{\partial}{\partial u} - \frac{\partial}{\partial v} \right) \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right)$$

[using (3) and (5)]

$$\begin{aligned}
or, r &= \frac{\partial}{\partial u} \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) - \frac{\partial}{\partial v} \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \\
&= \frac{\partial^2 z}{\partial u^2} - 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2} \quad \dots \quad (6) \\
t &= \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left( \frac{\partial z}{\partial y} \right) = \left( \frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) \left( \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) \\
&\quad [using (3) and (5)]
\end{aligned}$$

$$\begin{aligned}
or, \quad t &= \frac{\partial}{\partial u} \left( \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) + \frac{\partial}{\partial v} \left( \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) \\
&= \frac{\partial^2 z}{\partial u^2} + 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2} \quad \dots \quad (7)
\end{aligned}$$

Using (6) and (7) in (1), the required canonical form is

$$\begin{aligned}
\left( \frac{\partial^2 z}{\partial u^2} - 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2} \right) - \left( \frac{\partial^2 z}{\partial u^2} + 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2} \right) &= 0 \\
\Rightarrow \frac{\partial^2 z}{\partial u \partial v} &= 0
\end{aligned}$$

2. Reduce the equation

$$(n-1)^2 \left( \frac{\partial^2 z}{\partial x^2} \right) - y^{2n} \left( \frac{\partial^2 z}{\partial y^2} \right) = ny^{2n-1} \left( \frac{\partial z}{\partial y} \right)$$

to canonical form and find its general solution.

**Solution**

The given equation is

$$\begin{aligned}
(n-1)^2 \left( \frac{\partial^2 z}{\partial x^2} \right) - y^{2n} \left( \frac{\partial^2 z}{\partial y^2} \right) &= ny^{2n-1} \left( \frac{\partial z}{\partial y} \right) \\
\Rightarrow (n-1)^2 r - y^{2n} t - ny^{2n-1} q &= 0 \quad \dots \quad (1)
\end{aligned}$$

Comparing it with

$$Rr + Ss + Tt + f(x, y, z, p, q) = 0$$

we get

$$R = (n-1)^2, S = 0, T = -y^{2n}$$

The  $\lambda$  –quadratic equation is

$$\begin{aligned}
R\lambda^2 + S\lambda + T &= 0 \\
\Rightarrow (n-1)^2 \lambda^2 - y^{2n} &= 0 \\
\Rightarrow \lambda &= \pm (n-1)^{-1} y^n \\
\therefore \lambda_1 &= (n-1)^{-1} y^n, \quad \lambda_2 = -(n-1)^{-1} y^n
\end{aligned}$$

Next, the characteristic equations are

$$\begin{aligned}
\frac{dy}{dx} + \lambda_1 &= 0 \text{ and } \frac{dy}{dx} + \lambda_2 = 0 \\
\Rightarrow \frac{dy}{dx} + (n-1)^{-1} y^n &= 0 \text{ and } \frac{dy}{dx} - (n-1)^{-1} y^n = 0
\end{aligned}$$

Integration implies

$$x - y - n + 1 = c_1$$

and

$$x + y - n + 1 = c_2$$

In order to reduce (1) to canonical form, we take

$$\begin{aligned}
&\left. \begin{aligned} u &= x - y^{-n+1} \\ \text{and } v &= x + y^{-n+1} \end{aligned} \right\} \quad \dots \quad (2)
\end{aligned}$$

$$\therefore p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}$$

and

$$q = \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} = (n-1)y^{-n} \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right)$$

Therefore,

$$\left. \begin{aligned} \frac{\partial}{\partial x} &= \frac{\partial}{\partial u} + \frac{\partial}{\partial v} \\ \text{and } \frac{\partial}{\partial y} &= \frac{\partial}{\partial u} - \frac{\partial}{\partial v} \end{aligned} \right\}$$

$$r = \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left( \frac{\partial z}{\partial x} \right) = \left( \frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) \left( \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right)$$

$$= \frac{\partial^2 z}{\partial u^2} + 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}$$

$$t = \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left( \frac{\partial z}{\partial y} \right) = \frac{\partial}{\partial y} \left( (n-1)y^{-n} \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \right)$$

$$= -n(n-1)y^{-n-1} \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) + (n-1)y^{-n} \frac{\partial}{\partial y} \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right)$$

$$= -n(n-1)y^{-n-1} \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) + (n-1)y^{-n} \left[ \frac{\partial}{\partial u} \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \frac{\partial u}{\partial y} + \frac{\partial}{\partial v} \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \frac{\partial v}{\partial y} \right]$$

$$= -n(n-1)y^{-n-1} \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) + (n-1)^2 y^{-2n} \left( \frac{\partial^2 z}{\partial u^2} - 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2} \right)$$

Substituting the values of  $r, t, q$  in (1) and simplifying we get

$$\frac{\partial^2 z}{\partial u \partial v} = 0 \quad \dots \quad (3)$$

which is the required canonical form of (1).

Now, integrating (3) w.r.t.  $v$  we get

$$\frac{\partial z}{\partial u} = F(u),$$

$F$  is an arbitrary function of  $u$ .

Again, integrating w.r.t.  $u$ , we get

$$z = \int F(u) du + G(v)$$

$$z = H(u) + G(v)$$

Where  $H(u) = \int F(u) du$  and  $H(v)$  are arbitrary functions. Using (2), the required solution of the given equation is

$$z = H(x - y^{-n+1}) + G(x + y^{-n+1})$$

### Example

Reduce  $3 \left( \frac{\partial^2 z}{\partial x^2} \right) + 10 \left( \frac{\partial^2 z}{\partial x \partial y} \right) + 3 \left( \frac{\partial^2 z}{\partial y^2} \right) = 0$  to canonical form and hence solve it.

$$\text{Ans: } z = f(y - 3x) + g(3y - x).$$

### Working Rule for Reducing a Parabolic Equation to it's Canonical Form

#### Step 1

Let the given equation

$$Rr + Ss + Tt + f(x, y, z, p, q) = 0 \quad \dots \quad (1)$$

be parabolic i.e.

$$S^2 - 4RT = 0.$$

### Step 2

Write  $\lambda$  –quadratic equation

$$R\lambda^2 + S\lambda + T = 0 \quad \dots \quad (2)$$

Let  $\lambda_1, \lambda_2$  be two roots of (2).

### Step 3

Write the characteristic equation for  $\lambda = \lambda_1$ , i.e.  $\frac{dy}{dx} + \lambda_1 = 0$ . Solving it, we get

$$f_1(x, y) = c_1 \quad \dots \quad (3)$$

$c_1$  being an arbitrary constant.

### Step 4

Choose  $u = f_1(x, y)$  and  $v = f_2(x, y) \quad \dots \quad (4)$

where  $f_2$  is an arbitrary function of  $x$  and  $y$  and is independent of  $f_1(x, y)$ . For this, verify that

$$J = \frac{\partial(u, v)}{\partial(x, y)} = \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} \neq 0 \quad \dots \quad (5)$$

### Step 5

Using relation (4) find  $p, q, r, s, t$  in terms of  $u$  and  $v$ .

### Step 6

Substituting the values of  $p, q, r, s, t$  obtained in step (5) and simplifying we get the following canonical form of (1) as

$$\frac{\partial^2 z}{\partial u^2} = \phi\left(u, v, z, \frac{\partial z}{\partial u}, \frac{\partial z}{\partial v}\right).$$

Or,

$$\frac{\partial^2 z}{\partial v^2} = \psi\left(u, v, z, \frac{\partial z}{\partial u}, \frac{\partial z}{\partial v}\right).$$

### Examples

1. Reduce the equation  $\frac{\partial^2 z}{\partial x^2} + 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 0$  to Canonical form and solve it.

Solution

The given PDE is

$$\begin{aligned} \frac{\partial^2 z}{\partial x^2} + 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} &= 0 \\ \Rightarrow r + 2s + t &= 0 \quad \dots \quad (1) \end{aligned}$$

Comparing (1) with

$$Rr + Ss + Tt + f(x, y, z, p, q) = 0,$$

we have

$$\begin{aligned} R &= 1, S = 2, T = 1. \\ \therefore S^2 - 4RT &= 2^2 - 4.1.1 = 0 \end{aligned}$$

Therefore, (1) is parabolic.

The  $\lambda$ -quadratic equation is, therefore

$$\lambda^2 + 2\lambda + 1 = 0 \Rightarrow \lambda = -1, -1 \text{ (equal roots)}$$

The corresponding characteristic equation is

$$\begin{aligned} \frac{dy}{dx} - 1 &= 0 \\ \Rightarrow dx - dy &= 0. \end{aligned}$$



Integration implies

$$x - y = c_1,$$

$c_1$  being an arbitrary constant.

Choose  $u = x - y$  and  $v = x + y$  ... (2)

Here,  $v = x + y$  is chosen in such a way that  $u$  and  $v$  are independent functions.

Now,

$$J = \frac{\partial(u, v)}{\partial(x, y)} = \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} = 1.1 - (-1).1 \neq 0.$$

Now,

$$p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \quad \dots \quad (3)$$

[using (2)]

$$q = \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} = -\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \quad \dots \quad (4)$$

[using (3)]

From (2) and (3)

$$\left. \begin{aligned} \frac{\partial}{\partial x} &= \frac{\partial}{\partial u} + \frac{\partial}{\partial v} \\ \frac{\partial}{\partial y} &= -\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \end{aligned} \right\} \quad \dots \quad (5)$$

$$\therefore r = \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left( \frac{\partial z}{\partial x} \right)$$

$$= \left( \frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) \left( \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right),$$

[using (3) and (5)]

$$= \frac{\partial}{\partial u} \left( \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) + \frac{\partial}{\partial v} \left( \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right)$$

$$= \frac{\partial^2 z}{\partial u^2} + 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2} \quad \dots \quad (6)$$

$$t = \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left( \frac{\partial z}{\partial y} \right) = \left( -\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) \left( -\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right)$$

[using (4) and (5)]

$$= -\frac{\partial}{\partial u} \left( -\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) + \frac{\partial}{\partial v} \left( -\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right)$$

And

$$s = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left( \frac{\partial z}{\partial y} \right) = \left( \frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) \left( -\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right)$$

[using (4) and (5)]

$$= \frac{\partial}{\partial u} \left( -\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) + \frac{\partial}{\partial v} \left( -\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right)$$

$$= -\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2} \quad \dots \quad (8)$$

Using (6), (7) and (8) in (1) the required canonical form is

$$\frac{\partial^2 z}{\partial v^2} = 0, i.e. \quad \frac{\partial}{\partial v} \left( \frac{\partial z}{\partial v} \right) = 0 \quad \dots \quad (9)$$

**To find the solution**

From (9)

$$\frac{\partial}{\partial v} \left( \frac{\partial z}{\partial v} \right) = 0$$

Integration implies

$$\frac{\partial z}{\partial v} = \phi(u) \quad \dots \quad (10)$$

$\phi$  being an arbitrary function.

Again, integrating we get

$$\begin{aligned} z &= \int \phi(u) dv + \psi(u) \\ &= v\phi(u) + \psi(u) \\ \Rightarrow z &= (x+y)\phi(x-y) + \psi(x-y) \end{aligned}$$

which is the required general solution,  $\phi$  and  $\psi$  being arbitrary functions.

2. Reduce the equation

$$y^2 \left( \frac{\partial^2 z}{\partial x^2} \right) - 2xy \left( \frac{\partial^2 z}{\partial x \partial y} \right) + x^2 \left( \frac{\partial^2 z}{\partial y^2} \right) = \frac{y^2}{x} \frac{\partial z}{\partial x} + \frac{x^2}{y} \frac{\partial z}{\partial y}$$

to canonical form and leave solve it.

**Solution**

The given PDE is

$$\begin{aligned} y^2 \left( \frac{\partial^2 z}{\partial x^2} \right) - 2xy \left( \frac{\partial^2 z}{\partial x \partial y} \right) + x^2 \left( \frac{\partial^2 z}{\partial y^2} \right) &= \frac{y^2}{x} \frac{\partial z}{\partial x} + \frac{x^2}{y} \frac{\partial z}{\partial y} \\ \Rightarrow y^2 r - 2xys + x^2 t - \frac{y^2}{x} p - \frac{x^2}{y} q &= 0 \quad \dots \quad (1) \end{aligned}$$

Comparing it with

$$Rr + Ss + Tt + f(x, y, z, p, q) = 0$$

we have  $R = y^2, S = -2xy, T = x^2$ .

So,

$$S^2 - 4RT = 4x^2y^2 - 4x^2y^2 = 0.$$

Therefore, (1) is parabolic.

the  $\lambda$ -quadratic equation is

$$\begin{aligned} R\lambda^2 + S\lambda + T &= 0 \\ \Rightarrow y^2\lambda^2 - 2xy\lambda + x^2 &= 0 \\ \Rightarrow (y\lambda - x)^2 &= 0 \\ \therefore \lambda &= \frac{x}{y}, \frac{x}{y} \end{aligned}$$

The corresponding characteristic equation is

$$\begin{aligned} \frac{dy}{dx} + \frac{x}{y} &= 0 \\ \Rightarrow xdx + ydy &= 0 \end{aligned}$$

Integration implies

$$\frac{x^2}{2} + \frac{y^2}{2} = c$$

$c$  being arbitrary constant.

Choose

$$\left. \begin{aligned} u &= \frac{x^2}{2} + \frac{y^2}{2} \\ v &= \frac{x^2}{2} - \frac{y^2}{2} \end{aligned} \right\} \quad \dots \quad (2)$$

Here  $v = \frac{x^2}{2} - \frac{y^2}{2}$  is chosen in such a way that  $u$  and  $v$  are independent as verified below.

$$J = \frac{\partial(u, v)}{\partial(x, y)} = \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} = -2xy \neq 0.$$

Now,

$$p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} = x \left( \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) \quad \dots \quad (3)$$

[using (2)]

$$q = \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} = y \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \quad \dots \quad (4)$$

[using (2)]

$$\begin{aligned} \therefore r &= \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left( \frac{\partial z}{\partial x} \right) \\ &= \frac{\partial}{\partial x} \left\{ x \left( \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) \right\}, \\ &= \left( \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) + x \frac{\partial}{\partial x} \left( \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) \\ &\quad \text{[using (3)]} \\ &= \left( \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) + x \left[ \frac{\partial}{\partial u} \left( \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) \frac{\partial u}{\partial x} + \frac{\partial}{\partial v} \left( \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) \frac{\partial v}{\partial x} \right] \\ &= \left( \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) + x^2 \left( \frac{\partial^2 z}{\partial u^2} + 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2} \right) \quad \dots \quad (5) \end{aligned}$$

[using (3)]

$$\begin{aligned} t &= \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left\{ y \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \right\} \\ &= \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) + y \left\{ \frac{\partial}{\partial y} \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \right\} \\ &\quad \text{[using (4)]} \\ &= \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) + y \left\{ \frac{\partial}{\partial u} \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \frac{\partial u}{\partial y} + \frac{\partial}{\partial v} \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \frac{\partial v}{\partial y} \right\} \\ &= \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) + y^2 \left( \frac{\partial^2 z}{\partial u^2} - 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2} \right) \quad \dots \quad (6) \end{aligned}$$

And

$$\begin{aligned} S &= \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left( \frac{\partial z}{\partial y} \right) \\ &= \frac{\partial}{\partial x} \left( y \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \right) \\ &= y \left[ \frac{\partial}{\partial u} \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \frac{\partial u}{\partial x} + \frac{\partial}{\partial v} \left( \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \frac{\partial v}{\partial x} \right] \\ &\therefore s = xy \left( \frac{\partial^2 z}{\partial u^2} - \frac{\partial^2 z}{\partial v^2} \right) \quad \dots \quad (7) \end{aligned}$$

Using (3), (4), (5), (6), and (7) in (1) and simplifying, we get

$$4x^2y^2\left(\frac{\partial^2 z}{\partial v^2}\right) = 0$$

$$\Rightarrow \frac{\partial^2 z}{\partial v^2} = 0 \quad \dots \quad (8)$$

which is the required Canonical form.

### To find the solution

From (8)

$$\frac{\partial^2 z}{\partial v^2} = 0$$

$$\Rightarrow \frac{\partial}{\partial v} \left( \frac{\partial z}{\partial v} \right) = 0$$

Integration implies

$$\frac{\partial z}{\partial v} = \phi(u) \quad \dots \quad (9)$$

$\phi$  is an arbitrary function.

Integrating (9) w.r.t.  $v$ , we get,

$$z = \int \phi(u) dv + \psi(u)$$

$\psi$  being arbitrary function.

$$\Rightarrow z = \phi(u) \int dv + \psi(u)$$

$$\Rightarrow z = v \phi(u) + \psi(u)$$

using (2), we get

$$z = \frac{x^2 - y^2}{2} \phi\left(\frac{x^2 + y^2}{2}\right) + \psi\left(\frac{x^2 + y^2}{2}\right)$$

3. Reduce  $x^2r + 2xys + y^2t = 0$  to canonical form and hence solve.

Ans: canonical form  $\frac{\partial^2 z}{\partial v^2} = 0$  and general solution is

$z = y\phi\left(\frac{y}{x}\right) + \psi\left(\frac{y}{x}\right)$ , where  $\phi$  and  $\psi$  are arbitrary functions.

### Working Rule for Reducing an Elliptic Equation to it's Canonical Form:

#### Step 1

Let the given equation

$$Rr + Ss + Tt + f(x, y, z, p, q) = 0 \quad \dots \quad (1)$$

be elliptic so that

$$S^2 - 4RT < 0$$

#### Step 2

Write the  $\lambda$ -quadratic equation

$$R\lambda^2 + S\lambda + T = 0 \quad \dots \quad (2)$$

Let  $\lambda_1, \lambda_2$  be the complex roots of (2)

#### Step 3

Then, the corresponding characteristic equations are

$$\frac{dy}{dx} + \lambda_1 = 0 \text{ and } \frac{dy}{dx} + \lambda_2 = 0$$

Solving these, we get the solution of the form

$$\left. \begin{aligned} f_1(x, y) + if_2(x, y) &= c_1 \\ f_1(x, y) - if_2(x, y) &= c_2 \end{aligned} \right\} \quad \dots \quad (3)$$

#### Step 4

Choose  $u = f_1(x, y) + if_2(x, y)$  and  $v = f_1(x, y) - if_2(x, y)$ .

Let  $\alpha$  and  $\beta$  be two new real variables such that

$u = \alpha + i\beta$  and  $v = \alpha - i\beta$  so that

$$\left. \begin{aligned} \alpha &= f_1(x, y) \\ \beta &= f_2(x, y) \end{aligned} \right\} \quad \dots \quad (4)$$

#### Step 5

Using relations (4), find  $p, q, r, s$  and  $t$  in terms of  $\alpha$  and  $\beta$ .

#### Step 6

Substituting these values of  $p, q, r, s, t$  and relations (4) in (1) and simplifying we will get the following canonical form of (1)

$$\frac{\partial^2 z}{\partial \alpha^2} + \frac{\partial^2 z}{\partial \beta^2} = \phi \left( \alpha, \beta, z, \frac{\partial z}{\partial \alpha}, \frac{\partial z}{\partial \beta} \right).$$

**Example** Reduce  $\frac{\partial^2 z}{\partial x^2} + x^2 \frac{\partial^2 z}{\partial y^2} = 0$  to canonical form

#### Solution

The given PDE is

$$\frac{\partial^2 z}{\partial x^2} + x^2 \frac{\partial^2 z}{\partial y^2} = 0 \quad \dots \quad (1)$$

Comparing it with

$$Rr + Ss + Tt + f(x, y, z, p, q) = 0,$$

we have  $R = 1, S = 0, T = x^2$ .

$$\therefore S^2 - 4RT = 0 - 4 \cdot 1 \cdot x^2 = -4x^2 < 0, x \neq 0.$$

Therefore, (1) is elliptic.

The  $\lambda$  -quadratic equation is

$$\begin{aligned} R\lambda^2 + S\lambda + T &= 0 \\ \Rightarrow \lambda^2 + x^2 &= 0 \\ \therefore \lambda &= ix, -ix \end{aligned}$$

So, the corresponding characteristic equations are

$$\frac{dy}{dx} + ix = 0 \text{ and } \frac{dy}{dx} - ix = 0$$

Integration implies

$$y + i\frac{x^2}{2} = c_1 \text{ and } y - i\frac{x^2}{2} = c_2$$

Choose

$$u = y + i\frac{x^2}{2} = \alpha + i\beta \text{ and } v = y - i\frac{x^2}{2} = \alpha - i\beta$$

where,

$$\left. \begin{aligned} \alpha &= y \\ \beta &= \frac{x^2}{2} \end{aligned} \right\} \quad \dots \quad (2)$$

are independent real variables.

Now,

$$p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial \alpha} \cdot \frac{\partial \alpha}{\partial x} + \frac{\partial z}{\partial \beta} \cdot \frac{\partial \beta}{\partial x}$$

$$= x \frac{\partial z}{\partial \beta} \quad \dots \quad (3)$$

[by (2)]

$$q = \frac{\partial z}{\partial y} = \frac{\partial z}{\partial \alpha} \cdot \frac{\partial \alpha}{\partial y} + \frac{\partial z}{\partial \beta} \cdot \frac{\partial \beta}{\partial y}$$

$$= \frac{\partial z}{\partial \alpha} \quad \dots \quad (4)$$

[by (2)]

$$r = \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left( \frac{\partial z}{\partial x} \right)$$

$$= \frac{\partial}{\partial x} \left( x \frac{\partial z}{\partial \beta} \right)$$

[by (3)]

$$= \frac{\partial z}{\partial \beta} + x \frac{\partial}{\partial x} \left( \frac{\partial z}{\partial \beta} \right)$$

$$= \frac{\partial z}{\partial \beta} + x \left[ \frac{\partial}{\partial \alpha} \left( \frac{\partial z}{\partial \beta} \right) \frac{\partial \alpha}{\partial x} + \frac{\partial}{\partial \beta} \left( \frac{\partial z}{\partial \beta} \right) \frac{\partial \beta}{\partial x} \right]$$

$$= \frac{\partial z}{\partial \beta} + x^2 \frac{\partial^2 z}{\partial \beta^2} \quad \dots \quad (5)$$

$$t = \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left( \frac{\partial z}{\partial y} \right)$$

$$= \frac{\partial}{\partial y} \left( \frac{\partial z}{\partial \alpha} \right)$$

[by (4)]

$$= \frac{\partial}{\partial \alpha} \left( \frac{\partial z}{\partial \alpha} \right), \text{ since } \alpha = y$$

$$= \frac{\partial^2 z}{\partial \alpha^2} \quad \dots \quad (6)$$

Using (3) and (6) in (1), the required canonical form is

$$\frac{\partial z}{\partial \beta} + x^2 \frac{\partial^2 z}{\partial \beta^2} + x^2 \frac{\partial^2 z}{\partial \alpha^2} = 0$$

$$\text{or, } x^2 \frac{\partial^2 z}{\partial \alpha^2} + x^2 \frac{\partial^2 z}{\partial \beta^2} = - \frac{\partial z}{\partial \beta}$$

$$\text{or, } \frac{\partial^2 z}{\partial \alpha^2} + \frac{\partial^2 z}{\partial \beta^2} = - \frac{1}{x^2} \frac{\partial z}{\partial \beta}$$

$$\text{or, } \frac{\partial^2 z}{\partial \alpha^2} + \frac{\partial^2 z}{\partial \beta^2} = - \frac{1}{2\beta} \frac{\partial z}{\partial \beta},$$

$$\text{where, } \beta = \frac{x^2}{2} \Rightarrow x^2 = 2\beta.$$

Example      Reduce  $y^2 \frac{\partial^2 z}{\partial x^2} + x^2 \frac{\partial^2 z}{\partial y^2} = 0$  to canonical form.

$$\text{Ans: } \frac{\partial^2 z}{\partial \alpha^2} + \frac{\partial^2 z}{\partial \beta^2} + \frac{1}{2} \left( \frac{1}{\alpha} \frac{\partial z}{\partial \alpha} + \frac{1}{\beta} \frac{\partial z}{\partial \beta} \right) = 0.$$

### Monge's Methods

The general form of PDE of second order is

$$f(x, y, z, p, q, r, s, t) = 0 \quad \dots \quad (1)$$

Monge's method of solving (1) is to find one or two integrals of the form

$$u = \phi(u) \quad \dots \quad (2)$$

where  $u$  and  $v$  known functions of  $x, y, z, p$  and  $q$  and  $\phi$  is an arbitrary function. That is, this method finds relations of the form (2) such that equation (1) is obtained from (2) by eliminating arbitrary function  $\phi$ .

The relations of the form (2) is called an intermediate integral of (1). Every equation (1) does not necessarily have an intermediate integral.

The most general PDE having (2) as an intermediate integrals are of the following forms:

$$\left. \begin{aligned} Rr + Ss + Tt &= V \\ Rr + Ss + Tt + U(rt - S^2) &= v \end{aligned} \right\} \quad \dots \quad (3)$$

where  $R, S, T, V, U$  are function of  $x, y, z, p, q$ .

**Monge's Method of Integrating  $Rr + Ss + Tt = V$**

Given equation

$$Rr + Ss + Tt = V \quad \dots \quad (1)$$

where  $R, S, T, V$  are functions of  $x, y, z, p$  and  $q$ .

We have

$$\left. \begin{aligned} p &= \frac{\partial z}{\partial x} \\ q &= \frac{\partial z}{\partial y} \\ r &= \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left( \frac{\partial z}{\partial x} \right) = \frac{\partial p}{\partial x} \\ t &= \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left( \frac{\partial z}{\partial y} \right) = \frac{\partial q}{\partial y} \\ s &= \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left( \frac{\partial z}{\partial y} \right) = \frac{\partial q}{\partial x} \\ \text{or, } s &= \frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y} \left( \frac{\partial z}{\partial x} \right) = \frac{\partial p}{\partial y} \end{aligned} \right\} \quad \dots \quad (2)$$

Now,

$$\begin{aligned} dp &= \frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy \\ \therefore dp &= r dx + s dy \quad \dots \quad (3) \end{aligned}$$

and

$$\begin{aligned} dq &= \frac{\partial q}{\partial x} dx + \frac{\partial q}{\partial y} dy \\ \therefore dq &= s dx + t dy \quad \dots \quad (4) \end{aligned}$$

From (3) and (4)

$$\left. \begin{aligned} r &= \frac{(dp - s dy)}{dx} \\ \text{and } t &= \frac{(dq - s dx)}{dy} \end{aligned} \right\} \quad \dots \quad (5)$$

Putting these values of  $r$  and  $t$  in (1) we get

$$\begin{aligned} R \frac{(dp - s dy)}{dx} + Ss + T \frac{(dq - s dx)}{dy} &= V \\ \text{or, } R(dp - s dy)dy + Ss dx dy + T(dq - s dx)dx &= V dx dy \\ \text{or, } (R dp dy + T dq dx - V dx dy) - S\{R(dy)^2 - S dx dy + T(dx)^2\} &= 0 \quad \dots \quad (6) \end{aligned}$$

Any relation among  $x, y, z, p$  and  $q$  that satisfies (6) also satisfies the following two simultaneous equations.

$$Rdpdy + Tdqdx - Vdxdy = 0 \quad \dots \quad (7)$$

and

$$R(dy)^2 - Sdxdy + T(dx)^2 = 0 \quad \dots \quad (8)$$

The equations (7) and (8) are called Monge's subsidiary equations and the relations which satisfy these equations are called intermediate integrals.

Since equation (8) is quadratic, it can be factored into two equations say

$$dy - m_1 dx = 0 \quad \dots \quad (9)$$

and

$$dy - m_2 dx = 0 \quad \dots \quad (10)$$

### There are Two Possible Cases

#### Case I

Where  $m_1$  and  $m_2$  in (9) and (10) are distinct.

In this case (7) and (9), if necessary, by using the relation

$$dz = p dx + q dy$$

will give two integrals

$$u_1 = a$$

and

$$v_1 = b$$

where  $a$  and  $b$  are arbitrary constants. These will give

$$u_1 = f_1(v_1) \quad \dots \quad (11)$$

$f_1$  being an arbitrary function. This is called an intermediate integral of (1).

Next, taking (7) and (10) as before, we get another intermediate integral of (1), say

$$u_2 = f_2(v_2) \quad \dots \quad (12)$$

$f_2$  being an arbitrary function.

Thus, we obtain two distinct intermediate integrals (11) and (12).

Solving (11) and (12) we find  $p$  and  $q$  internal of  $x, y$  and  $z$ . Now, substituting these value of  $p$  and  $q$  in the relation

$$dz = p dx + q dy \quad \dots \quad (13)$$

and integrating it, we get the required complete integral of (1).

#### Case II

#### When $m_1 = m_2$ i.e. (8) is a Perfect Square

In this case we get only one intermediate integral which is in Lagrange's form

$$Pp + Qq = R \quad \dots \quad (14)$$

Solving (1) using Lagrange's method, we get the required complete integral of (1).

### The PDEs of the Form

$$Rr + Ss + Tt = V$$

are Divided into Following Four Types.

#### Type I

Where  $Rr + Ss + Tt = V$  possesses two distinct intermediate integrals and both of them are used to the required solution.

#### Working Rule

1. Write down the given PDE in the standard form

$$Rr + Ss + Tt = V$$



2. Substitute the values of  $R, S, T, V$  in Monge's subsidiary equations

$$Rdpdy + Tdqdx - Vdxdy = 0 \quad \dots \quad (1)$$

and

$$R(dy)^2 - Sdxdy + T(dx)^2 = 0 \quad \dots \quad (2)$$

3. Factorize (1) into two distinct factors.

4. Using one of the factors (2) will lead to an intermediate integral and using next factor we get another intermediate integral.

5. Solve the intermediate integral obtained in step (4) and get the values of  $p$  and  $q$ .

6. Putting the values of  $p$  and  $q$  in the relation

$$dz = pdx + qdy$$

and integrating it we get the required general solution of given equation.

### Examples

1. Solve by Monge's Method  $r = a^2t$

#### solution

The given PDE is

$$r = a^2t$$

$$\text{or, } r - a^2t = 0$$

Comparing it with

$$Rr + Ss + Tt = V,$$

we have

$$R = 1, S = 0, T = -a^2, V = 0.$$

$\therefore$  Monge's subsidiary equations

$$Rdpdy + Tdqdx - Vdxdy = 0$$

and

$$R(dy)^2 - Sdxdy + T(dx)^2 = 0$$

imply

$$dpdy - a^2dqdx = 0 \quad \dots \quad (2)$$

and

$$(dy)^2 - a^2(dx)^2 = 0 \quad \dots \quad (3)$$

Equation (3) can be factorized as

$$(dy - adx)(dy + adx) = 0$$

Hence, the two systems to be taken are

$$dpdy - a^2dqdx = 0, \quad dy - adx = 0 \quad \dots \quad (4)$$

$$dpdy - a^2dqdx = 0, \quad dy + adx = 0 \quad \dots \quad (5)$$

Integrating second equation of (4), we get

$$y - ax = c_1 \quad \dots \quad (6)$$

Eliminating  $\frac{dy}{dx}$  between the equation of (4) we get

$$dp - adq = 0 \Rightarrow p - aq = c_2 \quad \dots \quad (7)$$

Hence, the intermediate integral corresponding to (4) is

$$p - aq = \phi_1(y - ax) \quad \dots \quad (8)$$

Similarly, another intermediate integral corresponding to (5) is

$$p + aq = \phi_2(y + ax) \quad \dots \quad (9)$$

Here,  $\phi_1$  and  $\phi_2$  are arbitrary functions.

Solving (8) and (9) for  $p$  and  $q$ , we get

$$p = \frac{1}{2}[\phi_2(y + ax) + \phi_1(y - ax)]$$

and

$$q = \frac{1}{2a}[\phi_2(y + ax) - \phi_1(y - ax)]$$

Putting these values of  $p$  and  $q$  in

$$dz = p dx + q dy,$$

we get

$$\begin{aligned} dz &= \frac{1}{2}[\phi_2(y + ax) + \phi_1(y - ax)]dx + \frac{1}{2a}[\phi_2(y + ax) - \phi_1(y - ax)]dy \\ &= \frac{1}{2a}\phi_2(y + ax)(dy + adx) - \frac{1}{2a}\phi_1(y - ax)(dy - adx) \end{aligned}$$

Integration implies

$$z = \psi_2(y + ax) + \psi_1(y - ax),$$

where  $\psi_1$  and  $\psi_2$  are arbitrary functions.

2. Solve  $r + (a + b)s + abt = xy$  by Monge's Method

**Solution**

The given PDE is

$$r + (a + b)s + abt = xy \quad \dots \quad (1)$$

Comparing it with

$$Rr + Ss + Tt = V \quad \dots \quad (2)$$

we get

$$R = 1, S = a + b, T = ab, V = xy \quad \dots \quad (3)$$

Monge's subsidiary equations are

$$R dp dy + T dq dx - V dx dy = 0$$

and

$$R(dy)^2 - S dx dy + T(dx)^2 = 0$$

Using (3), these equations reduce to

$$dp dy + abd q dx - xy dx dy = 0 \quad \dots \quad (4)$$

and

$$(dy)^2 - (a - b) dx dy + ab(dx)^2 = 0 \quad \dots \quad (5)$$

Factorizing (5) we get

$$(dy - b dx)(dy - a dx) = 0$$

So, the two systems to be considered are

$$dp dy + abd q dx - xy dx dy = 0, dy - b dx = 0 \quad \dots \quad (6)$$

and

$$dp dy + abd q dx - xy dx dy = 0, dy + b dx = 0 \quad \dots \quad (7)$$

Integrating second equation of (6), we get

$$y - bx = c_1 \quad \dots \quad (8)$$

Eliminating  $\frac{dy}{dx}$  between the equations of (6) we get

$$\begin{aligned} dp + adq - xy dx &= 0 \\ \text{or, } dp + adq - x(c_1 + bx) dx &= 0 \quad \dots \quad (9) \end{aligned}$$

Integration (9), we get

$$p + aq - \frac{c_1}{2}x^2 - \frac{b}{3}x^3 = c_2$$

$$\Rightarrow p + aq - \frac{x^2}{2}(y - bx) - \frac{b}{3}x^3 = c_2$$

[Using (8)]

$$\therefore p + aq - \frac{1}{2}x^2y + \frac{b}{6}x^3 = c_2 \quad \dots \quad (10)$$

Using (8) and (10), the first intermediate integral is

$$p + aq - \frac{1}{2}yx^2 + \frac{1}{6}bx^3 = \phi_1(y - bx) \quad \dots \quad (11)$$

Similarly, another intermediate integral corresponding to the system (7) is

$$p + bq - \frac{1}{2}yx^2 + \frac{1}{6}ax^3 = \phi_2(y - ax) \quad \dots \quad (12)$$

where  $\phi_1$  and  $\phi_2$  are arbitrary functions.

Solving (11) and (12) for  $p$  and  $q$ , we have

$$p = \frac{1}{2}x^2y - \frac{1}{6}(a + b)x^3 + (a - b)^{-1}[a\phi_2(y - ax) - b\phi_1(y - bx)]$$

and

$$q = \frac{1}{6}x^3 + (a - b)^{-1}[\phi_1(y - bx) - \phi_2(y - ax)]$$

Putting these values in

$$dz = p dx + q dy,$$

we get

$$dz = \frac{1}{2}x^2y dx - \frac{1}{6}(a + b)x^3 dx + (a - b)^{-1}[\phi_2(y - bx)dx - \phi_1(y - ax)dx] + \frac{1}{6}x^3 dy + (a - b)^{-1}[\phi_1(y - bx)dy - \phi_2(y - ax)dy]$$

$$\text{or, } dz = \frac{1}{6}(3x^3y dx + x^3 dy) - \frac{1}{6}(a + b)x^3 dx - (b - a)^{-1}[\phi_2(y - bx)dx - \phi_1(y - ax)dx] - (b - a)^{-1}[\phi_1(y - bx)dy - \phi_2(y - ax)dy]$$

$$\text{or, } dz = \frac{1}{6}d(x^3y) - \frac{1}{6}(a + b)x^3 dx - (b - a)^{-1}\phi_2(y - ax)(dy - adx) - (b - a)^{-1}\phi_1(y - bx)(dy - bdx)$$

$$\text{or, } dz = \frac{1}{6}d(x^3y) - \frac{1}{6}(a + b)x^3 dx + (b - a)^{-1}\phi_2(y - ax)d(y - ax) - (b - a)^{-1}\phi_1(y - bx)d(y - bx)$$

Integration implies

$$z = \frac{1}{6}x^3y - \frac{1}{24}(a + b)x^4 + \psi_2(y - ax) + \psi_1(y - bx)$$

where  $\psi_1$  and  $\psi_2$  are arbitrary functions.

### Examples

Solve by Monge's Method

$$1. (r - t)xy - s(x^2 - y^2) = qx - py$$

$$\text{Ans: } z = \psi_1\left(\frac{y}{x}\right) + \psi_2(x^2 + y^2),$$

$\psi_1, \psi_2$  being arbitrary functions

$$2. (r - s)x = (t - s)y$$

$$\text{Ans: } \frac{z}{x + y} = \psi_2(x + y) + \psi_1\left(\frac{y}{x}\right),$$

$\psi_1, \psi_2$  being arbitrary functions.

3. Solve by Monge's method

$$(r - t)xy - s(x^2 - y^2) = qx - py$$

$$\text{Ans: } z = \psi_1(y/x) + \psi_2(x^2 + y^2),$$

$\psi_1, \psi_2$  are arbitrary functions.

4. Solve by Monge's method

$$(r - s)x = (t - s)y$$

$$\text{Ans: } \frac{z}{x + y} = \psi_2(x + y) + \psi_1(y/x),$$

where  $\psi_1, \psi_2$  are arbitrary functions.

## Type II

### When the Given Equation

$$Rr + Ss + Tt = V$$

Leads to Two Distinct Intermediate Integrals and Only One is Used to Get the Required Solution

### Working Rule

1. Write the given equation in the standard equation

$$Rr + Ss + Tt = V$$

2. Substitute the value of  $R, S, T$  and  $V$  in Monge's subsidiary equation

$$Rdpdy + Tdqdx - Vdxdy = 0 \quad \dots (1)$$

and

$$R(dy)^2 - Sdxdy + T(dx)^2 = 0 \quad \dots (2)$$

3. Factorize (1) into two distinct factors.

4. Take one of the factors in step (3) and use (2) to get an intermediate integral. We do not find second intermediate integral as in type I.

5. Rewrite the intermediate integral of the step 4 in the form of Lagrange's for  $Pp + Qq = R$ . Solving it with usual method we get the required general solution.

### Examples

1. Solve by Monge's Method

$$(r - s)y + (s - t)x + q - p = 0$$

### Solution

The given PDE is

$$\begin{aligned} (r - s)y + (s - t)x + q - p &= 0 \\ \Rightarrow (r - s)y + (s - t)x &= p - q \quad \dots (1) \end{aligned}$$

Comparing it with

$$Rr + Ss + Tt = V,$$

we get

$$R = y, S = x - y, T = -x, V = p - q$$

Using these values in Lagrange's subsidiary equations

$$Rdpdy + Tdqdx - Vdxdy = 0$$

and

$$R(dy)^2 - Sdxdy + T(dx)^2 = 0$$

we get

$$ydpdy - xdqdx + (q - p)dxdy = 0 \quad \dots (2)$$

and

$$y(dy)^2 - (x - y)dxdy - x(dx)^2 = 0 \quad \dots (3)$$

Equation (3) can be written as

$$\begin{aligned}
(dy + dx)(ydy - xdx) &= 0 \\
\therefore dy + dx &= 0 \\
\Rightarrow dy &= -dx \quad \dots \quad (4)
\end{aligned}$$

and

$$ydy - xdx = 0 \quad \dots \quad (5)$$

Using (4), (2) becomes

$$\begin{aligned}
-ydpdx - xdqdx + qdx(-dx) - pxdy &= 0 \\
\Rightarrow ydp - xdq + qdx + pdy &= 0 \\
\Rightarrow ydp + pdy + (xdq + qdx) &= 0 \\
\Rightarrow d(y p) + d(x q) &= 0
\end{aligned}$$

Integration implies

$$y p + x q = c_1 \quad \dots \quad (6)$$

Integrating (4), we get

$$x + y = c_2 \quad \dots \quad (7)$$

Here  $c_1, c_2$  are arbitrary constants.

From (6) and (7) an intermediate integral is

$$y p + x q = f(x + y) \quad \dots \quad (8)$$

which is of the  $Pp + Qq = R$ .

So, it's Lagrange's subsidiary equations are

$$\frac{dx}{y} = \frac{dy}{x} = \frac{dz}{f(x + y)} \quad \dots \quad (9)$$

From first two ratios of (9), we have

$$\begin{aligned}
\frac{dx}{y} &= \frac{dy}{x} \Rightarrow xdx - ydy = 0 \\
\Rightarrow 2xdx - 2ydy &= 0
\end{aligned}$$

Integration implies

$$x^2 - y^2 = a^2 \quad \dots \quad (10)$$

$a$  being an arbitrary constant.

Taking first and third ratios of (9), we have

$$\begin{aligned}
\frac{dx}{y} &= \frac{dz}{f(x + y)} \\
\Rightarrow \frac{dx}{\sqrt{x^2 - a^2}} &= \frac{dz}{f(x + \sqrt{x^2 - a^2})} \\
&\quad [\text{Using (10)}] \\
\Rightarrow dz &= f\left(x + \sqrt{x^2 - a^2}\right) \sqrt{x^2 - a^2} dx \quad \dots \quad (11)
\end{aligned}$$

Putting  $x + \sqrt{x^2 - a^2} = v$  so that

$$\begin{aligned}
\left[1 + \frac{x}{\sqrt{x^2 - a^2}}\right] dx &= dv \quad \dots \quad (12) \\
\Rightarrow \left[\frac{x + \sqrt{x^2 - a^2}}{\sqrt{x^2 - a^2}}\right] dx &= dv \\
\Rightarrow \left[\frac{v}{\sqrt{x^2 - a^2}}\right] dx &= dv \\
\Rightarrow \frac{dx}{\sqrt{x^2 - a^2}} &= \frac{dv}{v}
\end{aligned}$$

$\therefore$  (11) reduce to

$$dz = \frac{1}{v} f(v) dv$$

$$\Rightarrow dz - \frac{1}{v} f(v) dv = 0$$

Integration implies

$$z - F(v) = b$$

$$\Rightarrow z - F(x + \sqrt{x^2 - a^2}) = b$$

$$\Rightarrow z - F(x + y) = b, \text{ using (10) } \dots (13)$$

From (10 and (13) the required general solution is

$$z - F(x + y) = G(x^2 - y^2)$$

$$\text{or, } z = F(x + y) + G(x^2 - y^2)$$

where  $F$  and  $G$  are arbitrary functions.

2. Solve  $rq^2 - 2pqs + p^2t = pt - qs$  by Monge's Method

**Solution**

The given PDE is

$$rq^2 - 2pqs + p^2t = pt - qs$$

$$q^2r - q(2ps - 1)s + p(p - 1)t = 0 \quad \dots (1)$$

Comparing (1) with

$$Rr + Ss + Tt + f(x, y, z, p, q) = 0,$$

we get

$$R = q^2, S = -q(2p - 1), T = p(p - 1), V = 0.$$

Using these values, Monge's subsidiary equations

$$Rdpdy + Tdqdx - Vdxdy = 0$$

and

$$R(dy)^2 - Sdxdy + T(dx)^2 = 0$$

reduce to

$$q^2dpdy + p(p - 1)dqdx = 0 \quad \dots (2)$$

and

$$q^2(dy)^2 + q(2p - 1)dxdp + p(p - 1)(dx)^2 = 0 \quad \dots (3)$$

Equation (3) implies

$$qdy + pdx = 0 \Rightarrow qdy = -pdx \quad \dots (4)$$

And

$$qdy + (p - 1)dx = 0 \quad \dots (5)$$

For the required solution, we consider (4) only.

Since

$$dz = pdx + qdy$$

equation (4) reduces to

$$dz = 0$$

$$\Rightarrow z = c_1 \quad \dots (6)$$

From (2)

$$(qdp)(qdy) + (p - 1)dq(pdx) = 0$$

$$\Rightarrow qdp(-pdx) + (p - 1)dq(pdx) = 0$$

$$[\text{using (4)}]$$

$$\Rightarrow qdp - (p - 1)dq = 0$$

$$\Rightarrow \frac{1}{p - 1}dp - \frac{1}{q}dq = 0$$

Integration implies

$$\log(p-1) - \log q = \log c_2 \Rightarrow \frac{p-1}{q} = c_2 \quad \dots \quad (7)$$

From (6) and (7), an intermediate integral of (1) is

$$\frac{p-1}{q} = f(z)$$

$$p - qf(z) = 1 \quad \dots \quad (8)$$

which is of Lagrange's form

$$Pp + Qq = R.$$

Lagrange's A.E. are

$$\frac{dx}{1} = \frac{dy}{-f(z)} = \frac{dz}{1} \quad \dots \quad (9)$$

From first and last ratios, we get

$$dx - dz = 0$$

Integration implies

$$x - z = c_2 \quad \dots \quad (10)$$

From last two fractions of (9), we have

$$dy - f(z)dz = 0$$

Integration implies

$$y - F(z) = c_3 \quad \dots \quad (11)$$

where  $F(z) = \int f(z) dz$ .

From (10) and (11), the required solution is

$$y - F(z) = G(x - z)$$

$$\Rightarrow y = F(z) + G(x - z)$$

where  $F$  and  $G$  are arbitrary functions.

3. Solve  $xr + (x+y)s + yt + p + q = 0$  by Monge's Method

$$\text{Ans: } z = F(x-y) + G\left(\frac{x}{y}\right).$$

### Type III

**When the Given Equation  $Rr + Ss + Tt = V$  Leads to Two Identical Intermediate Integrals**

#### Working Rule

1. Write the given equation in standard form  $Rr + Ss + Tt = V$
2. Substitute the values of  $R, S, T$  and  $V$  in Monge's subsidiary equations

$$Rdpdy + Tdqdx - Vdxdy = 0 \quad \dots \quad (1)$$

and

$$R(dy)^2 - Sdxdy + T(dx)^2 = 0 \quad \dots \quad (2)$$

3. RHS of (2) reduces to a perfect square and hence, it gives only one factor.
4. Start with the only one factor of step 3 and use (2) to get an intermediate integral.
5. Rewrite the intermediate integral in step 4 in the form  $Pp + Qq = R$  and use Lagrange's method to obtain the required general solution of the given equation.

#### Examples

1. Solve  $(1+q)^2r - 2(1+p+q+pq)s + (1+p)^2t = 0$

#### Solution

The given PDE is

$$(1+q)^2r - 2(1+p+q+pq)s + (1+p)^2t = 0 \quad \dots \quad (1)$$

Comparing it with

$$Rr + Ss + Tt = V,$$

we get

$$R = (1 + q)^2, S = -2(1 + p + q + pq), T = (1 + p)^2, \quad V = 0 \quad \dots \quad (2)$$

Monge's subsidiary equations are

$$Rdpdy + Tdqdx - Vdxdy = 0 \quad \dots \quad (3)$$

and

$$R(dy)^2 - Sdxdy + T(dx)^2 = 0 \quad \dots \quad (4)$$

Using (2) in (3) and (4) we get

$$(1 + q)^2 dpdy + (1 + p)^2 dqdx = 0 \quad \dots \quad (5)$$

and

$$(1 + q)^2 (dy)^2 + 2(1 + p + q + pq) dxdy + (1 + p)^2 (dx)^2 = 0 \quad \dots \quad (6)$$

Equation (6) can be written as

$$\begin{aligned} [(1 + q)dy + (1 + p)dx]^2 &= 0 \\ \Rightarrow (1 + q)dy + (1 + p)dx &= 0 \\ \Rightarrow (1 + q)dy &= -(1 + p)dx \quad \dots \quad (7) \end{aligned}$$

Using (7), (5) reduces to

$$\begin{aligned} (1 + q)dp\{(1 + q)dy\} - (1 + p)dq\{-(1 + p)dx\} &= 0 \quad \dots \quad (8) \\ \Rightarrow (1 + q)dp - (1 + p)dq &= 0 \\ \Rightarrow \frac{dp}{1 + p} - \frac{dq}{1 + q} &= 0 \end{aligned}$$

Integration implies

$$\begin{aligned} \ln(1 + p) - \ln(1 + q) &= \ln c_1 \\ \Rightarrow \frac{1 + p}{1 + q} &= c_1 \quad \dots \quad (9) \end{aligned}$$

$c_1$  being an arbitrary constant.

From (7)

$$\begin{aligned} dx + dy + pdx + qdy &= 0 \\ \text{or, } dx + dy + dz &= 0, \quad \because dz = pdx + qdy \end{aligned}$$

Integration implies

$$x + y + z = c_2 \quad \dots \quad (10)$$

$c_2$  being an arbitrary constant.

From (9) and (10), an intermediate integral of (1) is

$$\begin{aligned} 1 + p &= (1 + q)F(x + y + z) \\ \Rightarrow p - qF(x + y + z) &= F(x + y + z) - 1 \quad \dots \quad (11) \end{aligned}$$

which is of the form  $Pp + Qq = R$ .

The Lagrange's A.E. are

$$\frac{dx}{1} = \frac{dy}{-F(x + y + z)} = \frac{dz}{F(x + y + z) - 1} \quad \dots \quad (12)$$

Choosing 1, 1, 1 as multipliers, each ratio of (12) equals to  $\frac{dx+dy+dz}{0}$ .

$$\therefore dx + dy + dz = 0$$

Integration implies

$$x + y + z = c_2 \quad \dots \quad (13)$$

Taking first two ratios of (12), we have

$$\frac{dx}{1} = \frac{dy}{-F(x + y + z)}$$



$$\Rightarrow dx = -\frac{dy}{F(c_2)}, \text{ using (13)}$$

$$\Rightarrow dy + F(c_2)dx = 0$$

Integration implies

$$y + xF(c_2) = c_3$$

$$\Rightarrow y + xF(x + y + z) = c_3 \quad \dots \quad (14)$$

From (13) and (14), the required general solution is

$$y + xF(x + y + z) = G(x + y + z)$$

where  $F, G$  are arbitrary functions.

2. Solve  $y^2r - 2ys + t = p + 6y$  by Monge's Method.

**Solution**

The given PDE is

$$y^2r - 2ys + t = p + 6y \quad \dots \quad (1)$$

Comparing it with

$$Rr + Ss + Tt = V$$

we have,

$$R = y^2, S = -2y, T = 1, V = p + 6y \quad \dots \quad (2)$$

Monge's subsidiary equations are

$$Rdpdy + Tdqdx - Vdxdy = 0$$

and

$$R(dy)^2 - Sdxdy + T(dx)^2 = 0$$

Using (2) we get

$$y^2dpdy + dqdx - (p + 6y)dxdy = 0 \quad \dots \quad (3)$$

and

$$y^2(dy)^2 + 2dxdy + (dx)^2 = 0$$

$$\Rightarrow (ydy + dx)^2 = 0 \quad \dots \quad (4)$$

From (4)

$$ydy + dx = 0$$

$$dx = -ydy \quad \dots \quad (5)$$

Putting  $dx = -ydy$  from (5) in (1), we get

$$y^2dpdy + dq(-ydy) - (p + 6y)dyx(-ydy) = 0$$

$$\text{or, } ydp - dq + (p + 6y)dy = 0$$

$$\text{or, } (ydp + pdy) - dq + 6ydy = 0$$

Integration implies

$$yp - q + 3y^2 = c_1 \quad \dots \quad (6)$$

$c_1$  being arbitrary constant.

Integrating (5), we get

$$\frac{y^2}{2} + x = \frac{c_2}{2}$$

$$\Rightarrow y^2 + 2x = c_2 \quad \dots \quad (7)$$

From (6) and (7), an intermediate integral of (1) is

$$yp - q + 3y^2 = F(y^2 + 2x)$$

$$\text{or, } yp - q = F(y^2 + 2x) - 3y^2 \quad \dots \quad (8)$$

which is of the form  $Pp + Qq = R$ , where the Lagrange's A.E. are

$$\frac{dx}{y} = \frac{dy}{-1} = \frac{dz}{F(y^2 + 2x) - 3y^2} \quad \dots \quad (9)$$

From first two fractions, we get

$$\begin{aligned} \frac{dx}{y} &= \frac{dy}{-1} \\ \Rightarrow ydy + dx &= 0 \\ \Rightarrow 2ydy + 2dx &= 0 \end{aligned}$$

Integration implies

$$y^2 + 2x = c_3 \quad \dots \quad (10)$$

Taking last two ratios of (9), we get

$$\begin{aligned} \frac{dy}{-1} &= \frac{dz}{F(y^2 + 2x) - 3y^2} \\ \Rightarrow \frac{dy}{-1} &= \frac{dz}{F(c_3) - 3y^2}, \text{ using (10)} \\ \Rightarrow dz + [F(c_3) - 3y^2]dy &= 0 \end{aligned}$$

Integration implies

$$\begin{aligned} z + yF(c_3) - y^3 &= c_4 \\ \text{or, } z + yF(y^2 + 2x) - y^3 &= c_4 \quad \dots \quad (11) \end{aligned}$$

From (10) and (11), the required general solution is

$$z + yF(y^2 + 2x) - y^3 = G(y^2 + 2x)$$

where  $F$  and  $G$  are arbitrary functions.

3. Solve  $x^2r + zxys + y^2t = 0$  by Monge's Method.

$$\text{Ans: } z = yG\left(\frac{y}{x}\right) - H\left(\frac{y}{x}\right).$$

#### Type IV

**When  $Rr + Ss + Tt = V$  has no Intermediate Integrals**

#### Examples

1. Solve  $(q + 1)s = (p + q)t$  by Monge's Method

#### Solution

The given PDE is

$$\begin{aligned} (q + 1)s &= (p + q)t \\ \Rightarrow (q + 1)s - (p + q)t &= 0 \quad \dots \quad (1) \end{aligned}$$

Comparing it with  $Rr + Ss + Tt = V$ , we have

$$R = 0, S = q + 1, T = -(p + 1), V = 0.$$

Monge's subsidiary equations are

$$Rdpdy + Tdqdx - Vdxdy = 0$$

and

$$R(dy)^2 - Sdxdy + T(dx)^2 = 0.$$

These imply

$$-(p + 1)dqdx = 0 \quad \dots \quad (2)$$

and

$$-(q + 1)dxdy - (p + 1)(dx)^2 = 0 \quad \dots \quad (3)$$

From (2)

$$\begin{aligned} -(p + 1)dqdx &= 0 \\ \Rightarrow dq &= 0 \quad \dots \quad (4) \end{aligned}$$

and from (3), we get (dividing by  $dx$ )

$$\begin{aligned}
(q+1)dy + (p+1)dx &= 0 \quad \dots \quad (5) \\
\Rightarrow dx + dy + pdx + qdy &= 0 \\
\Rightarrow dx + dy + dz &= 0 \quad \because dz = pdx + qdy
\end{aligned}$$

Integrating it, we get

$$x + y + z = c_1 \quad \dots \quad (6)$$

Integrating (4), we get

$$q = c_2 \quad \dots \quad (7)$$

From (6) and (7), an intermediate integral of (1) is

$$\begin{aligned}
q &= f(x + y + z) \\
\Rightarrow \frac{dz}{dy} &= f(x + y + z)
\end{aligned}$$

Integrating it w.r.t.  $y$ , we get

$$z = F(x + y + z) + G(x),$$

where  $F(x + y + z) = \int f(x + y + z) dy$  and  $G(x)$  are arbitrary functions.

Hence, the required general solution of (1) is

$$z = F(x + y + z) + G(x).$$

2. Solve  $pq = x(ps - qr)$  by Monge's Method.

**Solution**

The given PDE is

$$\begin{aligned}
pq &= x(ps - qr) \\
\Rightarrow xqr - xps + 0.t &= -pq \quad \dots \quad (1)
\end{aligned}$$

Comparing it with  $Rr + Ss + Tt = V$ , we get

$$R = xq, S = xp, T = 0, V = -pq \quad \dots \quad (2)$$

Monge's subsidiary equations are

$$Rdpdy + Tdqdx - Vdxdy = 0$$

and

$$R(dy)^2 - Sdxdy + T(dx)^2 = 0.$$

Using (2), we get

$$xqdpdy + pqdxdy = 0 \quad \dots \quad (3)$$

and

$$xq(dy)^2 + xp dxdy = 0 \quad \dots \quad (4)$$

From (3)

$$\begin{aligned}
xqdpdy + pqdxdy &= 0 \\
\Rightarrow qdy[xdp + pdx] &= 0 \\
\Rightarrow xdp + pdx &= 0 \quad \dots \quad (5)
\end{aligned}$$

From (4)

$$\begin{aligned}
xq(dy)^2 + xp dxdy &= 0 \\
\Rightarrow xdy[qdy + pdx] &= 0 \\
\Rightarrow qdy + pdx &= 0 \quad \dots \quad (6) \\
\Rightarrow dz = 0, \text{ since } dz &= pdx + qdy
\end{aligned}$$

Integration implies

$$z = c_1 \quad \dots \quad (7)$$

From (5), by integration we get

$$\begin{aligned}
xdp + pdx &= 0 \\
\Rightarrow d(xp) &= 0
\end{aligned}$$

$$\Rightarrow xp = c_2 \quad \dots \quad (8)$$

From (7) and (8), an integral of (1) is

$$\begin{aligned} xp &= f(z) \\ \text{or, } x \frac{\partial z}{\partial x} &= f(z) \\ \text{or, } \frac{1}{f(z)} \partial z &= \frac{1}{x} \partial x \end{aligned}$$

Integrating, we get

$$\begin{aligned} \ln f(z) &= \ln x + G(y) \\ \Rightarrow F(z) &= \ln x + G(y) \end{aligned}$$

Where,  $F(z) = \ln f(z)$ ,  $G(y)$  are arbitrary functions.

Therefore, the required general solution of (1) is

$$F(z) = \ln x + G(y).$$

4. Solve  $pt - qs = q^3$  by Monge's Method

$$\text{Ans: } y = xz - F(z) + G(x).$$

### Monge's Method of Solving

$$\mathbf{Rr + Ss + Tt + U(rt - s^2) = V}$$

The given equation is

$$Rr + Ss + Tt + U(rt - s^2) = V \quad \dots \quad (1)$$

we have

$$dp = \frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy = r dx + s dy$$

and

$$dq = \frac{\partial q}{\partial x} dx + \frac{\partial q}{\partial y} dy = s dx + t dy$$

From these two, we get

$$r = \frac{dp - s dy}{dx}$$

and

$$t = \frac{dq - s dx}{dy}.$$

Putting these values of  $r$  and  $t$  in (1), we get

$$(Rdpdy + Tdqdx - Udpdq - Vdxdy) - s\{R(dy)^2 - Sdxdy + T(dx)^2 + Udpdx + Udqdy\} = 0$$

Hence, the usual Monge's Subsidiary equations are

$$L \equiv Rdpdy + Tdqdx + Udpdq - Vdxdy = 0 \quad \dots \quad (2)$$

and

$$M \equiv R(dy)^2 - Sdxdy + T(dx)^2 + Udpdx + Udqdy = 0 \quad \dots \quad (3)$$

We cannot factorize  $M$  as in the previous case because of the presence of terms  $Udpdx + Udqdy$ .

Hence, let us factorize

$$M + \lambda L,$$

Where  $\lambda$  is some multiplier to be determined. Now, we have

$$M + \lambda L \equiv R(dy)^2 + T(dx)^2 - (S - \lambda V)dxdy + Vdpdx + Udqdy + \lambda Rdpdy + \lambda Tdqdx + \lambda Udpdq = 0 \quad \dots \quad (4)$$

Factoring LHS of (4), let  $k$  and  $m$  be constants such that

$$M + \lambda L \equiv (Rdy + mTdx + kUdp) \left( dy + \frac{1}{m} dx + \frac{\lambda}{k} dp \right) = 0 \quad \dots \quad (5)$$

Comparing coefficients in (4) and (5) we get

$$\frac{R}{m} + mT = -(S + \lambda V) \quad \dots \quad (6)$$

$$k = m \text{ and } \frac{R\lambda}{k} = V \quad \dots \quad (7)$$

From two relations of (7), we get

$$m = \frac{R\lambda}{U}.$$

Putting this value of  $m$  in (6) and simplifying we get

$$\lambda^2(UV + RT) + \lambda US + U^2 = 0 \quad \dots \quad (8)$$

which is quadratic in  $\lambda$ . Let  $\lambda_1$  and  $\lambda_2$  be its roots.

When  $\lambda = \lambda_1$ , equation (7) implies

$$\frac{R\lambda_1}{k} = U \Rightarrow k = \frac{R\lambda_1}{V} \Rightarrow m = \frac{R\lambda_1}{V}$$

Hence, (5) gives

$$\begin{aligned} & \left( Rdy + \frac{R\lambda_1}{U}Tdx + R\lambda_1dp \right) \left( dy + \frac{U}{R\lambda_1} + \frac{U}{R}dq \right) = 0 \\ \Rightarrow & (Udy + \lambda_1Tdx + \lambda_1Udp)(Udx + \lambda_1Rdy + \lambda_1Udq) = 0 \quad \dots \quad (9) \end{aligned}$$

Similarly, for  $\lambda = \lambda_2$ , (5) gives

$$(Udy + \lambda_2Tdx + \lambda_2Udp)(Udx + \lambda_2Rdy + \lambda_2Udq) = 0 \quad \dots \quad (10)$$

Here, we combine the first factor of (9) with the second factor of (10) and second factor of (9) is combined with the first factor of (10). i.e. the factors of (9) and (10) are combined in the following way:

$$\begin{cases} Udy + \lambda_1Tdx + \lambda_1Udp = 0 \\ Udx + \lambda_2Rdy + \lambda_2Udq = 0 \end{cases} \quad \dots \quad (11)$$

And

$$\begin{cases} Udy + \lambda_2Tdx + \lambda_2Udp = 0 \\ Udx + \lambda_1Rdy + \lambda_1Udq = 0 \end{cases} \quad \dots \quad (12)$$

Let equations (11) give two integrals

$$u_1 = c \text{ and } v_1 = d$$

so that one intermediate integral is

$$u_1 = f_1(v_1) \quad \dots \quad (13)$$

where,  $f_1$  is an arbitrary function.

Similarly (12) gives second intermediate integral

$$u_2 = f_2(v_2) \quad \dots \quad (14)$$

where,  $f_2$  is an arbitrary function.

We now solve (13) and (14) for  $p$  and  $q$  and substitute in

$$dz = pdx + qdy,$$

which when integrated gives the required general solution.

### **Type I**

#### **When the Roots of $\lambda$ – Quadratic are Identical (Equal)**

Example Solve  $5r + 6s + 3t + 2(rt - s^2) + 3 = 0$  by Monge's Method.

Solution

The given PDE is

$$\begin{aligned} 5r + 6s + 3t + 2(rt - s^2) + 3 &= 0 \\ 5r + 6s + 3t + 2(rt - s^2) &= -3 \quad \dots \quad (1) \end{aligned}$$

Comparing it with

$$Rr + Ss + Tt + U(rt - s^2) = V \quad \dots \quad (2)$$

we get,

$$R = 5, S = 6, T = 3, U = 2, V = -3 \quad \dots \quad (3)$$

The  $\lambda$ -quadratic equation is

$$\lambda^2(CV + TR) + \lambda SU + U^2 = 0$$

Using (3), it reduces to

$$9\lambda^2 + 12\lambda + 4 = 0$$

$$\Rightarrow (3\lambda + 2)^2 = 0$$

$$\therefore \lambda = \lambda_1 = \lambda_2 = -\frac{2}{3}$$

So, there is only one intermediate integral given by

$$Udy + \lambda_1 Tdx + \lambda_1 Udp = 0$$

and

$$Udx + \lambda_2 Rdy + \lambda_2 Udq = 0$$

Or,

$$2dy - \frac{2}{3} \cdot 3dx - \frac{2}{3} \cdot 2dp = 0$$

and

$$2dx - \frac{2}{3} \cdot 5dy - \frac{2}{3} \cdot 2dq = 0$$

Or,

$$3dy - 3dx - 2dp = 0$$

and

$$3dx - 5dy - 2dq = 0$$

Integrating, we get

$$\left. \begin{array}{l} 3y - 3x - 2p = c_1 \\ \text{and } 3x - 5y - 2q = c_2 \end{array} \right\} \quad \dots \quad (4)$$

Hence, an intermediate integral is

$$3y - 3x - 2p = f(3x - 5y - 2q) \quad \dots \quad (5)$$

where  $f$  is an arbitrary function.

Solving two equations of (4) for  $p$  and  $q$  we have

$$p = \frac{1}{2(3y - 3x - c_1)}$$

and

$$q = \frac{1}{2(3x - 5y - c_2)}$$

Putting these values of  $p$  and  $q$  in

$$dz = pdx + qdy,$$

we get,

$$\begin{aligned} dz &= \frac{1}{2(3y - 3x - c_1)} dx + \frac{1}{2(3x - 5y - c_2)} dy \\ \Rightarrow 2dz &= 3(ydx + xdy) - 3xdx - 5ydy - c_1dx - c_2dy \end{aligned}$$

Integration implies

$$2z = 3xy - \frac{3x^2}{2} - \frac{5y^2}{2} - 4x - c_1x - c_2y + c_3$$

which is the required complete integral where  $c_1, c_2, c_3$  are arbitrary constants.

**Examples**

Solve

$$\begin{aligned} 1) \quad & 2s + (rt - s^2) = 1 \\ 2) \quad & 3r + 4s + t + (rt - s^2) = 1 \end{aligned}$$

By Monge's Method.

$$\begin{aligned} \text{Ans: } (1) \quad & z = xy - c_1x - c_2y + c_3 \\ 2) \quad & z = 2xy - \frac{x^2}{2} - \frac{3y^2}{2} + c_1x + c_2y + c_3 \end{aligned}$$

## Type II

**When  $\lambda$ -Quadratic has Distinct Roots.**

### Example

Solve  $r + 4s + t + rt - s^2 = 2$  by Monge's Method

Solution

The given PDE is

$$r + 4s + t + rt - s^2 = 2 \quad \dots \quad (1)$$

Comparing it with

$$Rr + Ss + Tt + U(rt - s^2) = V$$

we get

$$R = 1, S = 4, T = 1, U = 1, V = 2 \quad \dots \quad (2)$$

The  $\lambda$ -quadratic equation is

$$\lambda^2(UV + RT) + \lambda VS + U^2 = 0 \quad \dots \quad (3)$$

Using (2), we get

$$\begin{aligned} 3\lambda^2 + \lambda + 1 &= 0 \\ \therefore \lambda_1 &= -1, \lambda_2 = -\frac{1}{3} \end{aligned}$$

Two integrals of (1) are given by

$$\left. \begin{aligned} Vdy + \lambda_1 Tdx + \lambda_1 Udp &= 0 \\ Vdx + \lambda_2 Rdy + \lambda_2 Udq &= 0 \end{aligned} \right\} \quad \dots \quad (4)$$

and

$$\left. \begin{aligned} Udy + \lambda_2 Tdx + \lambda_2 Udp &= 0 \\ Udx + \lambda_1 Rdy + \lambda_1 Udq &= 0 \end{aligned} \right\} \quad \dots \quad (5)$$

Using (2) and  $\lambda_1 = -1, \lambda_2 = -\frac{1}{3}$

Equation (4) and (5) reduce to

$$\left. \begin{aligned} dp + dx - dy &= 0 \\ dq + dy - 3dx &= 0 \end{aligned} \right\} \quad \dots \quad (6)$$

And

$$\left. \begin{aligned} dp + dx - 3dy &= 0 \\ dq + dy - dx &= 0 \end{aligned} \right\} \quad \dots \quad (7)$$

Integrating (6) and (7), we get

$$p + x - y = c_1, q + y - 3x = c_2 \quad \dots \quad (8)$$

and

$$p + x - 3y = c_3, q + y - x = c_4 \quad \dots \quad (9)$$

where,  $c_1, c_2, c_3, c_4$  are arbitrary constants.

From (8) and (9) the two intermediate integrals of (1) are

$$\left. \begin{aligned} p + x - y &= f(q + y - 3x) \\ \text{and } p + x - 3y &= g(q + y - x) \end{aligned} \right\} \quad \dots \quad (10)$$

Let

$$q + y - 3x = \alpha \quad \dots \quad (11)$$

and

$$q + y - x = \beta \quad \dots \quad (12)$$

$\therefore$  (10) reduces to

$$p + x - y = f(\alpha) \quad \dots \quad (13)$$

and

$$p + x - 3y = g(\beta) \quad (14)$$

Here  $\alpha$  and  $\beta$  are parameters.

Solving (11) and (12) for  $x$  and (13) and (14) for  $y$ , we get

$$x = \frac{\beta - \alpha}{2} \quad \dots \quad (15)$$

$$y = \frac{f(\alpha) - f(\beta)}{2} \quad \dots \quad (16)$$

From (13)

$$p = y - x + f(\alpha) \quad \dots \quad (17)$$

and from (12)

$$q = x - y + \beta \quad \dots \quad (18)$$

From (15) and (16) we have

$$\left. \begin{aligned} dx &= \frac{1}{2}(d\beta - d\alpha) \\ dy &= \frac{1}{2}[f'(\alpha) d\alpha - g'(\beta)d\beta] \end{aligned} \right\} \quad \dots \quad (19)$$

$$\therefore dz = p dx + q dy$$

$$\Rightarrow dz = [y - x + f(\alpha)]dx + [x - y + \beta]dy$$

$$= ydx + xdy - xdx - ydy + f(\alpha)dx + \beta dy$$

$$= d(xy) - xdx - ydy + \frac{1}{2}[f(\alpha)d\beta + \beta f'(\alpha)d\alpha] - \frac{1}{2}f(\alpha)d\alpha - \frac{1}{2}\beta g'(\beta)d\beta$$

$$2dz = 2d(xy) - 2xdx - 2ydy + d[\beta f(\alpha)] - f(\alpha)d\alpha - \beta g'(\beta)d\beta$$

Integration implies

$$2z = 2xy - x^2 - y^2 + \beta f(\alpha) - \int f(\alpha) d\alpha - \left[ \beta g(\beta) - \int 1 \cdot g(\beta) d\beta \right]$$

$$2z = 2xy - x^2 - y^2 + \beta[f(\alpha) - g(\beta)] - \int f(\alpha) d\alpha + \int g(\beta) d\beta$$

Let

$$\left. \begin{aligned} \int f(\alpha) d\alpha &= \phi(\alpha) \\ \int g(\beta) d\beta &= \psi(\beta) \end{aligned} \right\} \quad \dots \quad (20)$$

Therefore,

$$\left. \begin{aligned} f(\alpha) &= \phi'(\alpha) \\ g(\beta) &= \psi'(\beta) \end{aligned} \right\} \quad \dots \quad (21)$$

Using (15), (16), (20) and (21), we get

$$2x = \beta - \alpha, 2y = \phi'(\alpha) - \psi'(\beta),$$

$$2z = 2xy - x^2 - y^2 + \beta[\phi'(\alpha) - \psi'(\beta)] - \phi(\alpha) + \psi(\beta),$$

Which is the required solution in parametric form ( $\alpha$  and  $\beta$  being parameters) and  $\phi, \psi$  being arbitrary functions.

**Example** Solve by Monge's method:  $3s + rt - s^2 = 2$

$$\text{Ans: } z = xy + \beta[\phi'(\alpha) - \psi'(\beta)] - \phi(\alpha) + \psi(\beta),$$



where  $x = \alpha - \beta, y = f(\alpha) - g(\beta), \alpha = q - 2x, \beta = q - x$ .

**THE END**